

Research Paper

Magnetic field induced by convective flow in Europa's subsurface ocean

L. Šachl, J. Kvorka*, O. Čadek, J. Velínský

Department of Geophysics, Faculty of Mathematics and Physics, Charles University, V Holešovičkách 2, Praha 8, 18000, Czech Republic

ARTICLE INFO

Dataset link: <https://zenodo.org/records/13838989>

Keywords:

Europa
Magnetic fields
Ocean dynamics

ABSTRACT

Movements of Europa in Jupiter's magnetic field generate an induced magnetic field in the moon's interior. Its measurements by the Galileo space probe led to the discovery of Europa's subsurface ocean. However, interactions of the ocean flow with Jupiter's background magnetic field also generate the motionally induced electrical currents in the ocean and the corresponding ocean-induced magnetic field (OIMF), which has not yet been studied in detail. A single study estimated the $\text{OIMF} \leq 20 \text{ nT}$ using a simplified scaling relation. In this paper, we revisit this estimate using a physically consistent modeling setup. Based on the numerical simulations of ocean convection, we show that two modes can exist in Europa's ocean. Mode I is dominated by a prograde zonal flow at the equator with negligible radial and meridional flows. Mode II is characterized by Hadley-like meridional circulation cells in both hemispheres and a retrograde zonal flow at the equator. The scaling analysis based on our dataset strongly indicates that Mode II is appropriate for Europa's ocean with velocities around 0.3 m/s . We then calculate Europa's OIMF using a time-domain EM induction solver, which properly accounts for self-induction and diffusion of the magnetic field in the silicate and ice layers, and implicitly covers the full temporal spectrum. Our calculations suggest that even under the most favorable circumstances (150 km thick ocean with a conductivity of 18 S/m located under a 1 km thick ice layer) the magnitude of Europa's OIMF forced by the flow in Mode II is approximately 1 nT , at the lower bound of the sensitivity of the Europa Clipper measurements and more than one order of magnitude smaller than previously predicted. The discrepancy is primarily caused by a more sluggish ocean flow and a correct treatment of EM induction. Moreover, Europa's OIMF is affected by the electrical conductivity and thickness of ice and ocean, which we demonstrate in a parametric study.

1. Introduction

Electromagnetic (EM) methods played a crucial role in detecting the subsurface oceans on Europa and Ganymede (Zimmer et al., 2000; Khurana et al., 2002; Kivelson et al., 2002). Europa lacks an intrinsic magnetic field generated by dynamo action. However, as it passes through Jupiter's magnetic field (JMF; see Section 2.2.2 for details), its time variations induce a secondary EM field, which was foreseen by Colburn and Reynolds (1985) and then measured by the Galileo probe at a single frequency corresponding to the synodic period. Zimmer et al. (2000) studied the magnetic field perturbations measured by Galileo. They assumed a uniform JMF oscillating in time and a spherically symmetric conductive shell in Europa's interior. A comparison of the predicted induced dipolar field with the measurements from two flybys indicated that Europa contains a conductive layer. By adding measurements from the flyby on January 2000, Kivelson et al. (2000) confirmed that the magnetic field perturbations vary in time, which ruled out the hypothesis of Kivelson et al. (1997) that Europa has a permanent static dipole.

Despite these achievements, the potential of EM methods to probe the Jovian moons' interior has not yet been fully explored. Two space missions to the Jovian system (NASA's Europa Clipper and ESA's Juice) are expected to bring new data in the early 2030s, providing unprecedentedly detailed information on the structure and temporal variations of the magnetic field in the vicinity of Europa and Ganymede. Furthermore, the EM response at two different frequencies could constrain ocean thickness and conductivity simultaneously (Khurana et al., 2009; Seufert et al., 2011).

In addition to the passive EM response of an interior ocean to the external field variations, the flow of the electrically conductive saline fluids also interacts with the ambient Jovian magnetic field, giving rise to an ocean-induced magnetic field (OIMF) by the process of motional induction. The magnetic Lorentz force, acting on charge carriers in the direction perpendicular both to the flow velocity and the ambient magnetic field, drives electric currents, which further interact with the conductive medium through inductive and galvanic coupling. Surprisingly, little attention has been given to Europa's OIMF (EOIMF)

* Corresponding author.

E-mail address: kvorka@karel.troja.mff.cuni.cz (J. Kvorka).

so far. A notable exception is the study by Vance et al. (2021), who estimated the maximum amplitude of the radial EOIMF component to be 20 nT. However, their estimate was based on a simplified scaling relation in which they inserted typical values of the JMF, the electrical conductivity of the ocean, and the flow velocity according to the flow model of Soderlund (2019). The objective of this paper is to revisit their results using a physically consistent modeling approach.

The modeling of Earth's OIMF is well established. The pioneering papers of Cox et al. (1970), Sanford (1971), Sanford (1982), and Larsen and Sanford (1985) were followed by the forward modeling studies of Stephenson and Bryan (1992), Flosadóttir et al. (1997), Vivier et al. (2004), and Manoj et al. (2006). Interest among the EM community in the OIMF was further stimulated by Tyler et al. (2003), who managed to extract the tidally driven OIMF (TIMF) from the CHAMP satellite data. Grayver and Olsen (2019) and Sabaka et al. (2020) later provided more precise TIMF models derived from Swarm data. The OIMF within Earth's oceans was studied by Velínský et al. (2019, 2021). The TIMF was also used in the inverse modeling studies of Grayver et al. (2016, 2017), and Šachl et al. (2022, 2024) to probe the 1-D and 3-D electrical conductivity of the upper mantle. In the context of this study, the most influential papers are Velínský et al. (2018) and Šachl et al. (2019), which studied the numerical aspects of OIMF modeling. Our modeling approach is designed accordingly.

Similarly to OIMF modeling, Earth's ocean circulation modeling has a long history. Since the pioneering works of Bryan (1969), Semtner (1974), and Cox (1984), ocean general circulation models (OGCMs) have significantly evolved from eddy-parametrized to eddy-permitting and eddy-resolving models with tuned parameterizations to include the subgrid-scale processes. MOM, NEMO, MITgcm (and ECCO), FESOM2, ICON, MPIOM, ROMS, and POP2 are some of the well-established state-of-the-art OGCMs. The OGCMs are available in various configurations, including regional and global models, coupled climate models, and ocean-only simulations. The models differ in their vertical coordinates (geopotential z , isopycnal ρ and terrain-following σ or s coordinate and their hybrids) and discretization techniques (mostly finite difference/element/volume methods). However, the transition from the Earth's to planetary oceans is not so straightforward:

1. There is no free surface. The upper boundary of the subsurface ocean is in contact with a rigid ice shell.
2. The hydrostatic approximation, which is commonly applied, is invalid since the ocean is in a convective regime.
3. Similarly, the shallow-ocean approximation (Griffies, 2004) is invalid.
4. The ocean is global, and thus, the so-called Pole problem cannot be solved by relocating the coordinate-system singularity to the land.

Two different approaches are currently used to study the dynamics of subsurface oceans. Although both approaches solve the same governing equations, they differ in a number of details and are used to address different aspects of convective heat transfer. The first approach, hereafter referred to as DNS (Dimensionless Numerical Simulations), is based on the method developed to study the magnetic field generation in the Earth's liquid core (e.g., Christensen and Wicht, 2007). This approach usually assumes that the material parameters of the ocean are constant and the buoyancy forces are proportional to temperature. The system can then be described by three dimensionless parameters, called the Rayleigh (Ra), Ekman (Ek), and Prandtl (Pr) numbers. The behavior of the system under study is usually estimated from the statistical processing of large datasets obtained for computationally feasible values of Ra and Ek and their extrapolation to realistic conditions (e.g., Amit et al., 2020; Kvorka and Čadek, 2022; Kvorka and Čadek, 2024). The dynamics of Europa's ocean has been addressed in several DNS studies (Soderlund et al., 2014; Soderlund, 2019; Lemasquerier et al., 2023; Cabanes et al., 2024). Other studies have investigated the heat transfer in the ocean of Saturn's moon Titan and its link to ice thickness

variations (Amit et al., 2020; Kvorka and Čadek, 2022; Terra-Nova et al., 2023). Although the simulations in the latter studies were performed for a relatively thick ocean, the datasets, and scaling laws presented provide useful insight into subsurface ocean dynamics and can also be applied to Europa.

The second approach, hereafter referred to as GCM (Global Circulation Models), is based on the methodology developed to study convection in Earth's oceans (Marshall et al., 1997). In this approach, thermal and chemical density anomalies are consistently coupled through the equation of state, which is particularly important in studying small icy moons, such as Enceladus, where ocean salinity is likely to be relatively low (< 30 g/kg, Zeng and Jansen, 2021). While the DNS approach solves the problem on a global scale and extrapolates the results obtained by processing unrealistic datasets, the GCM approach uses a reduced geometry and incorporates the Reynolds averaging in the form of a prescribed viscosity model, thus avoiding the need to extrapolate the results (Visbeck et al., 1997). The GCM studies have mainly focused on Enceladus (e.g., Kang, 2022; Kang and Jansen, 2022) and Europa (Goodman et al., 2004; Goodman and Lenferink, 2012; Ashkenazy and Tziperman, 2021; Bire et al., 2022; Ashkenazy et al., 2023).

From the perspective of EOIMF, the flow patterns and speeds suggested for Europa's ocean in the studies cited above appear contradictory. Soderlund (2019) proposed that flows in Europa's ocean are dominated by Hadley-like circulation cells and three zonal jets with maximum speeds reaching approximately meters per second. In contrast, Ashkenazy and Tziperman (2021), Bire et al. (2022) and Cabanes et al. (2024) argued for a rapidly rotating ocean characterized by Taylor columns and two zonal jets in the equatorial region. Their models predict flow speeds ranging from less than 1 mm/s to 0.6 m/s. In both cases, the upper limit found for the flow speed is significantly higher than that obtained from the theoretical analysis of energetic constraints, which suggests that the mean flow speed in Europa's ocean is between 1.5 and 6 cm/s (Jansen et al., 2023). The objective of this paper is to clarify the discrepancy among these results and to reassess the velocity estimates to appraise the prospects of EOIMF measurements for future missions.

The paper is organized as follows. In Section 2, we explain the theoretical aspects of our modeling approach. Section 2.1 is devoted to ocean modeling. We introduce the governing equations, discuss the boundary conditions, and examine the dependence of a flow pattern on the Rayleigh, Ekman, and Prandtl numbers. EM modeling is addressed in Section 2.2. We briefly discuss the EM induction equation, its mathematical form, and its numerical implementation. We then focus on the JMF and discuss its dipolar characteristics and evolution in time. The section concludes with a list of thickness models and a representative conductivity model supported by the actual knowledge of Europa's interior. Section 3 is devoted to EOIMF calculated in the respective configurations. Finally, we summarize the results in Section 4.

2. Methodology

2.1. Ocean modeling

2.1.1. Governing equations and control parameters

In the present study, the flow in Europa's ocean is investigated using the DNS approach. We consider the convection of a Boussinesq fluid contained in a spherical shell rotating at a constant angular speed, ω , around the z -axis. The outer radius, r_o , and the inner radius, r_i , of the shell are chosen so that $r_i/r_o = 0.9$. The boundaries of the shell are impermeable, free-slip, and isothermal (for the discussion of the boundary conditions, see Kvorka and Čadek, 2022). The equations governing the heat and mass transport are expressed in the dimensionless form,

$$\nabla \cdot \mathbf{v} = 0, \quad (1)$$

$$\frac{1}{Pr} \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) + \frac{2}{Ek} \mathbf{e}_z \times \mathbf{v} = -\nabla p + \nabla^2 \mathbf{v} + Ra T g \mathbf{e}_r, \quad (2)$$

$$\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T = \nabla^2 T, \quad (3)$$

and solved using the pseudospectral method described in Kvorka and Čadek (2022). In the above equations, $\mathbf{v}(\mathbf{r}; t)$, $p(\mathbf{r}; t)$ and $T(\mathbf{r}; t)$ denote the dimensionless velocity, pressure and temperature, respectively, $g(r)$ is the gravitational acceleration ($g \propto r^{-2}$) and $\mathbf{e}_r$ is the radial unit vector. Convection is controlled by three dimensionless parameters: the Rayleigh, Ekman and Prandtl numbers,

$$Ek = \frac{\nu}{\omega D^2}, \quad Ra = \frac{\alpha g_o \Delta T D^3}{\nu \kappa}, \quad Pr = \frac{\nu}{\kappa}, \quad (4)$$

where ν is the kinematic viscosity, κ is the thermal diffusivity, g_o is the gravitational acceleration at the outer boundary of the ocean, D is the ocean thickness, ΔT is the superadiabatic temperature contrast across the ocean and α is the coefficient of thermal expansion.

The Ekman and Rayleigh numbers corresponding to the subsurface oceans of icy moons are of the order of $\lesssim 10^{-10}$ and $\gtrsim 10^{19}$, respectively (Soderlund, 2019). Simulations of ocean dynamics with such values of Ek and Ra are extremely expensive and out of reach for current computational methods. To avoid this difficulty, we perform a series of simulations for values of Ek and Ra that are numerically tractable and extrapolate the results to real conditions using the scaling laws (see Table A.3 for the list of numerical simulations). Unlike the previous studies (e.g., Gastine et al., 2015, 2016; Kvorka and Čadek, 2022) that mostly examined convection in relatively thick shells ($r_i/r_o \leq 0.8$), the simulations are carried out for the radius ratio of 0.9 that is more relevant to Europa (cf. Soderlund, 2019). Our goal is to determine the geometry of Europa's ocean circulation and estimate the typical flow speed.

Depending on the relative importance of the Coriolis, buoyancy, and inertia forces (i.e., depending on the values of parameters Ek , Ra , and Pr), one can identify three convection regimes: rapidly rotating, transitional, and non-rotating. Each regime is characterized by specific flow patterns, as discussed in, e.g., Gastine et al. (2016), Cheng et al. (2018), Soderlund (2019), Amit et al. (2020), Kvorka and Čadek (2022) and Cabanes et al. (2024). The transition from one regime to another can be described by two dimensionless numbers, the local convective Rossby number (Julien et al., 2012),

$$Ro_{loc} = Ra Ek^{8/5} Pr^{-3/5}, \quad (5)$$

and the convective Rossby number (Gilman, 1977),

$$Ro_c = Ra^{1/2} Ek Pr^{-1/2}. \quad (6)$$

The transition from the rapidly rotating regime to the transitional regime occurs at $Ro_{loc} \approx 1$ while the transition from the transitional regime to the non-rotating regime occurs at $Ro_c \approx 1$. In other words, the ocean is in the rapidly rotating regime if $Ro_{loc} < 1$, in the transitional regime if $Ro_{loc} > 1$ and $Ro_c < 1$, and in the non-rotating regime if $Ro_c > 1$. To calculate Ro_{loc} and Ro_c , we need to know the values of Pr , Ek and Ra . Using the same material parameters as Lemasquerier et al. (2023) and assuming that D is between 95 and 150 km (Casajus et al., 2021; Petricca et al., 2023), we obtain $Pr = 11.3$ and Ek ranging from $3.9 \cdot 10^{-12}$ to $9.5 \cdot 10^{-12}$. Unlike Pr and Ek , the Rayleigh number cannot be determined from Eq. (4) because the superadiabatic temperature contrast, ΔT , is unknown. As shown in detail in Appendix A, the problem can be circumvented by using the total heat flow, Q , in conjunction with the statistical analysis of the dataset obtained by solving Eqs. (1)–(3) for different values of Ek and Ra (Table A.3). Based on the model of tidal dissipation in Europa's interior by Běhouňková et al. (2021) and the study of Vance et al. (2018), we assume that Q is between 0.3 TW and 3.2 TW, which gives $Ra \in (1.3 \cdot 10^{21}, 2.7 \cdot 10^{22})$. Note that the range of Ra predicted here is compatible with but narrower than the range proposed by Soderlund (2019), who used the scaling laws developed by Gastine et al. (2015, 2016).

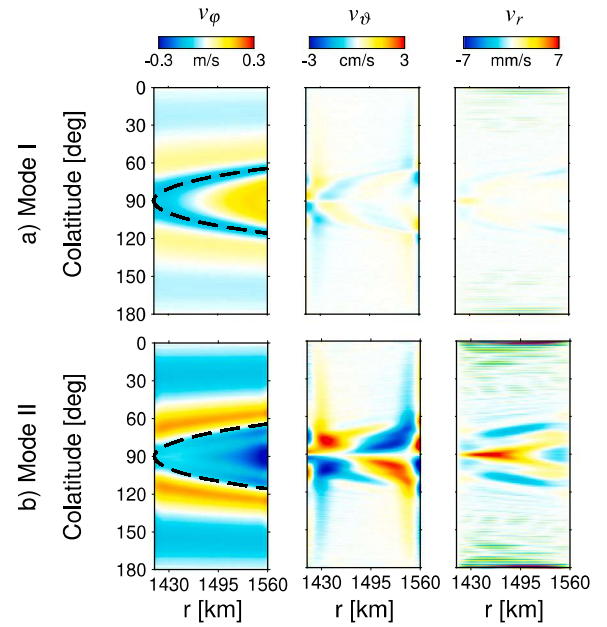


Fig. 1. (a) Flow velocity pattern typical of Mode I. The flow field [m/s] is obtained for model No. 31 and is scaled so that the mean speed, $U = 0.10$ m/s, corresponding to the ocean thickness of $D = 149$ km (model A in Fig. 8) and $Q = 0.3$ TW (the lowest value of Q considered in this study). (b) As in panel (a) but for Mode II and model No. 32. The velocity is scaled so that $U = 0.28$ m/s, corresponding to the heat flow of 3.2 TW. The velocity fields are averaged over at least 0.02 viscous diffusion times and the longitude. The dashed lines mark the position of the tangent cylinder (the cylinder with radius r_t whose axis is identical to the rotation axis).

2.1.2. Flow geometry

Substituting the above values of Pr , Ek and Ra in Eqs. (5) and (6), we obtain $Ro_{loc} > 720$ and $Ro_c < 0.2$, implying that Europa's ocean is in the transitional regime (see also Table A.1). Previous studies (Soderlund, 2019; Amit et al., 2020; Kvorka and Čadek, 2022) have shown that the structure of flow in the transitional regime is not uniform but varies depending on the relative importance of rotation (see, e.g., Figs. 5 and 6 in Kvorka and Čadek, 2022). The flow patterns can be divided into two basic categories, denoted here as Mode I and Mode II. The flow in Mode I is still significantly affected by rotation. It is characterized by two zonal jets of opposite directions in the equatorial region and relatively small radial and meridional velocities (see Fig. 1a). As the influence of the Coriolis force decreases relative to the buoyancy force, the equatorial zonal jet at the outer boundary changes from prograde to retrograde. The overturn is accompanied by the formation of meridional Hadley-like circulation cells in both hemispheres and upwellings concentrated in the equatorial region (Fig. 1b). This new flow pattern is denoted as Mode II. Motivated by the discussion in Aurnou et al. (2007), we distinguish Mode II from Mode I based on the sign of the zonal flow velocity at the equator on the outer boundary.

Analysis of our dataset suggests that the transition from Mode I to Mode II can be described by the dimensionless parameter

$$R_M = Ra Ek^{12/7} Pr^{-3/4}. \quad (7)$$

Note that for $Pr = 1$, R_M is identical to the parameter proposed by Gastine et al. (2016) to describe the transition from rotating to non-rotating convection. Parameter R_M is also rather similar to parameter $R_G^* = Ra Ek^{12/7} Pr^{-1}$ used by Kvorka and Čadek (2022) to analyze latitudinal variations of the heat flux at the upper boundary of the ocean. As shown in Fig. 2, the transition from Mode I to Mode II occurs at $R_M = 8$. Substitution of Pr , Ek , and Ra obtained for Europa in the previous section gives $R_M > 27$, meaning that Mode II appropriately represents the flow in Europa's ocean. In Section 3, the induced magnetic field is evaluated for models No. 31 and 32 from Table A.3 that can be

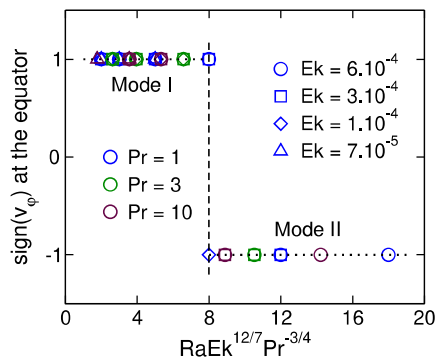


Fig. 2. The sign of the zonal equatorial jet on the outer boundary as a function of parameter $R_M = RaEk^{12/7} Pr^{-3/4}$. Shapes and colors of the symbols indicate the values of parameters Ek and Pr used in the numerical simulation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

considered as typical representatives of the flow geometry in Modes I and II, respectively.

2.1.3. Flow speed

In order to correctly predict the OIMF, we need to know not only the geometry of large-scale ocean circulation but also the flow speed. Vance et al. (2021) implemented the model of Soderlund (2019) under the assumption that the flow speeds obtained from numerical simulations are the same as those in the oceans of icy satellites. However, the speeds reported by Vance et al. (2021, $\lesssim 3$ m/s) are significantly larger than the values recently suggested for Europa's ocean by Jansen et al. (2023, 1.5–6 cm/s), Cabanes et al. (2024, $\lesssim 0.6$ m/s) and Kang (2024, ~ 1 cm/s). Cabanes et al. (2024) argue that this discrepancy arises from the assumption made by Vance et al. (2021) and that any physical quantity, including the velocity, should be rescaled to a realistic setting. Following this argument, we have developed a scaling law for the Rossby number, Ro , which can be interpreted as a dimensionless representation of the mean flow speed ($Ro = U D^{-1} \omega^{-1}$ where U is the root-mean-square speed). The derivation is detailed in Appendix A, and the scaling law is presented in Table A.2 together with the results of other studies. Our scaling law predicts U in the range of 8–28 cm/s, in rough agreement with the estimate of Lemasquerier et al. (2023) who reported a range of 3.5–22 cm/s.

The velocity in models No. 31 and 32 is scaled using the Rossby number obtained for ocean thicknesses in Fig. 8 and the total heat flow of 0.3 and 3.2 TW, respectively. The radial, meridional, and zonal velocities are approximately 0.7–1.7 cm/s, 0.9–3.4 cm/s, and 14–32 cm/s, respectively. The lowest values are obtained for the thinnest and least heated ocean, with the thickness of 95 km and the total heat flow of 0.3 TW (heat flux of 10 mW/m² at the outer ocean's boundary), while the highest values correspond to the ocean thickness of 149 km and the heat flow of 3.2 TW (125 mW/m²). For a detailed comparison with the results of previous studies, see Appendix A.

2.2. EM modeling

2.2.1. Governing equations

The electromagnetic induction (EMI) equation governs the generation of EOIMF $\mathbf{B}(\mathbf{r}; t)$ due to the interactions of saline fluid flows with the JMF in Europa's ocean,

$$\mu_0 \frac{\partial \mathbf{B}}{\partial t} + \nabla \times \left(\frac{1}{\sigma} \nabla \times \mathbf{B} \right) = \mu_0 \nabla \times \mathbf{L}, \quad (8)$$

where $\sigma(r)$ is the spherically symmetric electrical conductivity model of Europa's interior, and μ_0 is the magnetic permeability of free space. The magnetic component of the Lorentz force acting on a unit charge,

$$\mathbf{L}(\mathbf{r}; t) = \mathbf{v}(\mathbf{r}; t) \times \mathbf{B}_0(t), \quad (9)$$

depends on the fluid velocity $\mathbf{v}(\mathbf{r}; t)$, and the JMF $\mathbf{B}_0(t)$ (see Section 2.2.2 for the details). In contrast to Eqs. (1)–(3), all quantities in Eqs. (8) and (9) are expressed in SI units and in the local Europa spherical coordinate system, $\mathbf{r} = (r, \vartheta, \varphi)$.

By neglecting the interactions of the small, motionally induced magnetic field with the electromagnetic environment of Europa, the EOIMF becomes a potential field in the moon's exterior, with continuity condition applied across the surface,

$$\nabla^2 U = 0 \quad \text{for } r \geq R, \quad (10)$$

$$\mathbf{B} = -\nabla U \quad \text{at } r = R, \quad (11)$$

$$\lim_{r \rightarrow \infty} U = \mathcal{O}(r^{-2}), \quad (12)$$

where R is the radius of the surface. The magnetic field observable above the moon's surface is then the sum of the JMF $\mathbf{B}_0$, the magnetospheric field, the field induced in Europa's interior due to time-varying external field, and finally, the motionally induced contribution $\mathbf{B}$.

We solve the EMI equation (8) using the time-domain, spherical harmonic

-finite element approach (Velínský and Martinec, 2005; Velínský et al., 2019, 2021). The magnetic field is expanded in the series of vector spherical harmonic functions in the lateral coordinates (Varshalovich et al., 1988), and piecewise-linear 1-D finite elements in radius. For a spherically symmetric electrical conductivity profile $\sigma(r)$, the problem is decoupled in spherical harmonic degree and order. The forcing term $\mathbf{L}$ uses piecewise-constant radial parameterization and is non-zero only in the layers spanning the internal ocean of Europa. The Crank–Nicolson scheme is applied to the time-integration of the magnetic field from its initial condition. The method was originally developed for and validated on the modeling of OIMF on Earth (Šachl et al., 2019). We use homogeneous rather than stationary initial conditions to calculate the EOIMF due to shorter spin-up time t_{spin} . We tested that one year was sufficient for the spin-up process (not shown) and used this value in all presented calculations.

2.2.2. Jupiter's magnetic field

Europa's internal magnetic field is either weak or missing (Khurana et al., 1998; Kivelson et al., 2000). Therefore, we assume the ocean flows are interacting with the JMF. Additionally, we neglect the spatial variation of JMF across Europa's interior, given that Europa's size is negligible compared to its distance from Jupiter. We use the spherical-harmonic (SH) model of JMF of Connerney et al. (2022). Although small-scale JMF structures are present on the planet's surface, the JMF is predominantly dipolar at Europa's orbit since short wavelengths attenuate faster than long wavelengths (Figs. 3 and 4). Nevertheless, we also consider higher-order terms in our calculations. The JMF model of Connerney et al. (2022) is available up to SH degree 30, but we truncate the SH expansion at $j_{\max} = 13$ based on the author's recommendation.

The time variations of JMF experienced by Europa can be primarily attributed to the misalignment between Jupiter's dipole and rotation axes and Europa's movement along its orbit. The vector of the JMF horizontal component, given by the projection of JMF into Europa's equatorial plane (EEP), rotates in EEP, and its magnitude varies between approximately one-fifth and one-third of the vertical component that is perpendicular to EEP (Figs. 3 and 4). The secular variations of JMF are presumed to exist but are not provided by the model of Connerney et al. (2022).

The SH formalism is employed to calculate the JMF in Jupiter's spherical coordinates. However, it is necessary to express it in Europa's spherical coordinate system, where the EOIMF is calculated (see Section 2.2.1). Consequently, JMF components are transformed from spherical to Cartesian coordinates, rotated, and then transformed back to spherical coordinates.

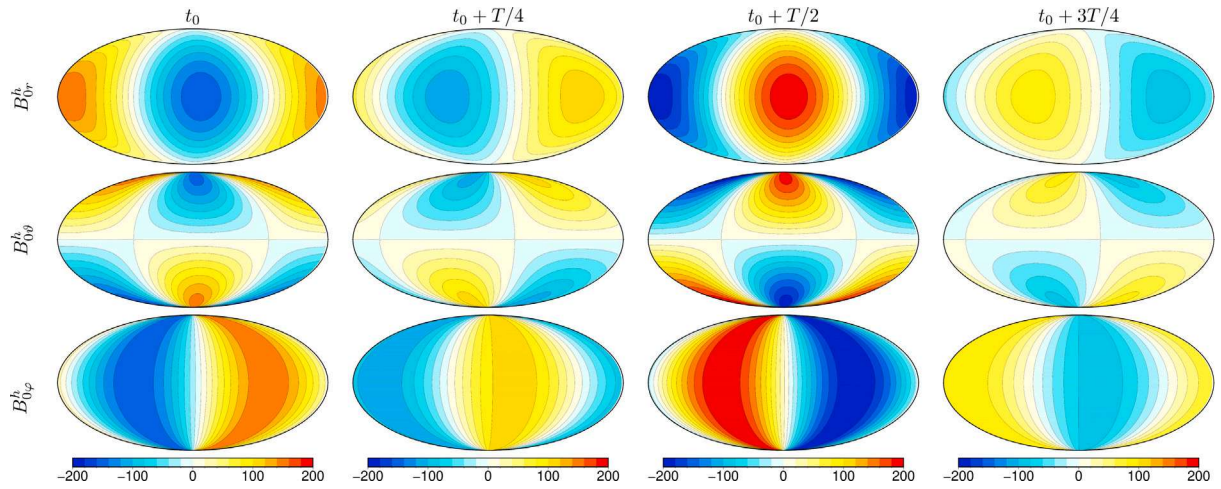


Fig. 3. Horizontal JMF [nT] on Europa's surface during the synodic period T starting from the initial time t_0 . The spherical harmonic expansion is truncated at $j_{\max} = 13$. The horizontal JMF is obtained by the projection of JMF into Europa's equatorial plane.

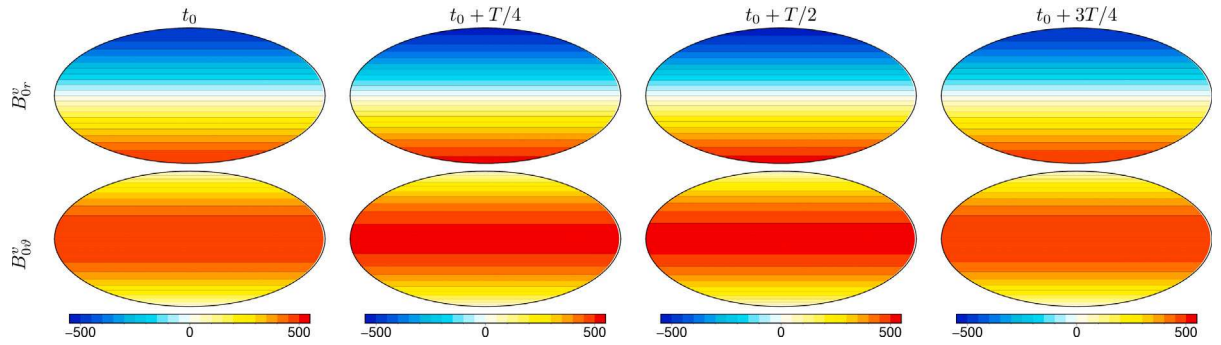


Fig. 4. As in Fig. 3 but for the vertical JMF. The zonal component is zero. The vertical JMF is defined as the part of JMF parallel to Europa's rotation axis.

2.2.3. Models of internal structure

We use a four-layer model of Europa's interior, consisting of the iron-nickel core, silicate mantle, ocean, and ice. We assign a constant electrical conductivity to each layer. The iron-nickel core is supposed to be highly conductive. We set the core conductivity to 10^3 S/m and the core outer radius to 431 km (Sohl et al., 2002; Kimura, 2024) in all calculations. Different core parameters should not significantly affect the results since the EOIMF signals attenuate in the mantle. We test several plausible models of Europa's interior regarding the conductivity and thickness of the mantle, ocean, and ice.

Let us begin with the thickness values. In recent years, considerable effort has been made to constrain the ice thickness H and the ocean thickness D . Billings and Kattenhorn (2005) even introduced the term “great thickness debate” for studies devoted to ice thickness. Casajus et al. (2021) analyzed the gravity field using revisited Galileo data to constrain the hydrosphere (ice + ocean) thickness. They estimated hydrosphere thickness to be 125–150 km. Petricca et al. (2023) presented the results of joint inversion of gravity and magnetic data. By combining two physical processes with different sensitivities, they could constrain both the ice and ocean thicknesses. In addition, they considered two induction amplitude values and three ocean chemical compositions. According to their calculations, the ice is 1–30 km thick, and the ocean is 80–155 km thick. Howell (2021) estimated the ice thickness to be 24.3 km, using a Bayesian analysis accounting for prior constraints based on geological analyses and geophysical parameters affecting possible solid-state convection. Lemasquerier et al. (2023) derived the ice thickness from ocean simulations. They predicted a 12 km thick ice shell for a thermal model dominated by tidal heating and a 24 km thick ice shell if radiogenic heating is dominant. Marusiak et al. (2021) estimated the ice thickness to be 5–30 km in their review paper.

Table 1

Models of Europa's hydrosphere considered in the study. The mean speed of the ocean's flow, U , is obtained by employing the scaling law LS24 (see Appendix A.2) and corresponds to the upper bound of the heat flux considered in this study ($q = 125$ mW/m² at the outer ocean's boundary).

Model	H [km]	D [km]	$H + D$ [km]	U [m/s]
A	1	149	150	0.28
B	1	124	125	0.25
C	30	120	150	0.25
D	30	95	125	0.22

For completeness, note that the Juno team's recent research (Levin et al., 2024) suggests that the minimum thickness of Europa's ice is larger than the upper limit from the review by Marusiak et al. (2021).

This paper builds upon the results of Casajus et al. (2021) and Petricca et al. (2023). We combine the estimate of the hydrosphere thickness of Casajus et al. (2021) with the estimate of the ice thickness of Petricca et al. (2023) to construct four thickness models that are summarized in Table 1. Models A–D encompass the range of ice and ocean thicknesses estimated by the abovementioned studies.

Regarding the electrical conductivity, Europa's mantle is predominantly composed of silicates, so Earth-like values of 10^{-2} S/m (Petricca et al., 2023) or 10^{-3} S/m (Marusiak et al., 2021) are expected. Additionally, we consider a nearly insulating mantle, using the value of 10^{-16} S/m consistent with the PlanetProfile outputs (Styczinski et al., 2023).

The conductivity of the ocean depends on its chemical composition. Europa's ocean dominated by NaCl, MgSO₄ or carbonates has been discussed in the literature. The respective upper conductivity limits are

18 S/m, 6 S/m and 6 S/m at 0 °C or 3.5 S/m at 5 °C (Hand and Chyba, 2007; Kobe and Sheehy, 1948). In the previously mentioned study of Petricca et al. (2023), the reconstructed conductivity values were between ~ 1 S/m and these conductivity limits. That agrees with Naseem et al. (2023), who considered the ocean conductivity of 1–10 S/m. Psarakis et al. (2024) conducted molecular dynamics simulations to derive electrical conductivity at high-pressure, low-temperature conditions. In Europa's ocean, the pressure of 0–200 MPa and the temperature of 270–275 K (Vance et al., 2023) are expected. For the more likely SPC/E water model and 10 wt % NaCl ions, the ocean conductivity can be estimated to be approximately 7.5 S/m. In contrast, Castillo-Rogez et al. (2022) estimated the ocean conductivity to be 0.5–0.7 S/m. There is no reliable information on lateral variations of Europa's electrical conductivity. For the saline ocean, it could be, in principle, related to the temperature and salinity obtained from the convection model using a suitable equation of state, but the uncertainties are presently too large to justify a computationally expensive 3-D EM modeling approach.

The ice is expected to be resistive. The conductivity values proposed by Petrenko and Schulson (1992) and Petricca et al. (2023) are 10^{-5} S/m and 10^{-6} S/m, respectively.

Consequently, we set the default values of mantle, ocean, and ice conductivity to $\sigma_m = 10^{-2}$ S/m, $\sigma_o = 10$ S/m and $\sigma_i = 10^{-5}$ S/m, respectively. Some of the alternative conductivity values cited above are considered in Section 3.4, where we study the EOIMF sensitivity to Europa's conductivity structure. For example, the default ocean conductivity of 10 S/m is a reasonable upper limit corresponding to a salty highly conductive ocean. But we also test the true upper conductivity limit of 18 S/m according to Hand and Chyba (2007) and address the possibility of a much more resistive ocean by using 1 S/m motivated by Castillo-Rogez et al. (2022).

3. Results

3.1. EOIMF symmetries

Before proceeding to the discussion of the predicted EOIMF models, we will demonstrate the elementary physics of motional induction and relate the geometric symmetries of EOIMF to the underlying ocean flow patterns. The flow velocity models averaged over time are axially symmetric around Europa's rotation axis (i.e., independent of the longitude) and mirror-symmetric about the equator (Fig. 1). This implies that the radial and zonal velocity components are symmetric, while the meridional component is antisymmetric about the equator: $v_r(r, \vartheta) = v_r(r, \pi - \vartheta)$, $v_\vartheta(r, \vartheta) = -v_\vartheta(r, \pi - \vartheta)$ and $v_\varphi(\vartheta) = v_\varphi(\pi - \vartheta)$, where ϑ denotes the colatitude. As for the horizontal JMF vector (Fig. 3), the meridional component is antisymmetric, and the radial and zonal components are symmetric, while in the case of the vertical JMF (Fig. 4), the meridional component is symmetric, and the radial component is antisymmetric.

Given the properties of the flow velocity and the JMF vector described above, one can expect that the resulting EOIMF vector will also have some level of symmetry. To investigate the EOIMF symmetries, we consider four combinations of flow velocity and JMF components (hereinafter referred to as Configurations 1–4, see Table 2), and for each of them, explore the symmetry properties of the corresponding EOIMF vector. In Table 2, B_0^h and B_0^v denote the horizontal and vertical components of JMF shown in Figs. 3 and 4, respectively, $v^P = v_r e_r + v_\vartheta e_\vartheta$ is the poloidal velocity field associated with convective cells, and $v^T = v_\varphi e_\varphi$ is the toroidal velocity field associated with zonal jets. The expressions for the Lorentz force, $\mathbf{L}$, and $\nabla \times \mathbf{L}$, corresponding to each configuration, can be found in Appendix B.1.

In interpreting the results presented in the last three columns of Table 2, it is essential to recognize that the vertical component of JMF is significantly larger than the horizontal component, and the circulation of Europa's ocean is dominated by the zonal flow; even in the case of Mode II, the poloidal flow which forms the convection cells in the meridional plane is an order of magnitude slower than the toroidal

Table 2

Summary of the surface EOIMF symmetries. S = symmetric field, A = antisymmetric field, N = neither symmetric nor antisymmetric, 0 = field equals zero. The surface EOIMF in Configuration 1 vanishes since it is a toroidal field.

Configuration	Jupiter's field	Flow	L_r	L_ϑ	L_φ	B_r	B_ϑ	B_φ
1	B_0^h	v^T	S	A	0	0	0	0
2	B_0^h	v^P	A	S	A	S	A	S
3	B_0^h	v^T	A	S	0	S	A	S
4	B_0^v	v^P	0	0	S	A	S	0
5	B_0	v	N	N	N	N	N	S

flow in the equatorial region. However, the interaction of the toroidal flow with the vertical JMF generates zero EOIMF, implying that the total EOIMF (denoted as Configuration 5 in Table 2) is mainly a superposition of Configurations 3 and 4 while Configuration 2 plays only a minor role. The zonal component of the total EOIMF is symmetric, but the radial and meridional components are neither symmetric nor antisymmetric since the symmetry properties of Configurations 2 and 3 differ from those of Configuration 4.

The EOIMF symmetries for Configurations 2–5 are illustrated in Fig. 5. Inspection of the figure shows that the EOIMF generated by convection cells and zonal jets exhibit comparable magnitudes even though the zonal jet speed is approximately one order of magnitude larger than the flow speed in convection cells. The EOIMF generated by convection cells manifests mainly in the radial and meridional components. The zonal component of EOIMF is mainly affected by the zonal flow, while the impact of convection cells is weak.

A remarkable feature demonstrated in Fig. 5 is the zonal stripes in the radial and meridional EOIMF components of Configuration 4. The origin of the zonal stripes is illustrated in Fig. 6. The Lorentz force generates zonal electrical currents that form a system of closed zonal loops, which in turn generate the EOIMF in the meridional plane. The total EOIMF is a superposition of partial EOIMFs from each loop, resulting in a system of narrow EOIMF stripes in the zonal direction. The EOIMF is constant in the zonal direction since both the vertical JMF and the velocity field are rotationally symmetric. However, the EOIMF exhibits relatively sharp sign changes in the meridional direction, reflecting the directional changes of the flow in the convection cells (upwellings and downwellings).

3.2. Comparison of EOIMF generated by Modes I and II

As discussed in Section 2.1, two flow configurations, Mode I and Mode II, may exist in Europa's subsurface ocean. Fig. 7 shows EOIMFs generated by Modes I and II for the thickness model A. We can identify the zonal stripes in the EOIMF generated by Mode II but not by Mode I since the convection cells in Mode I are much weaker or missing. The zonal components generated by the respective modes have different signs, indicating that the zonal jets flow in opposite directions in Modes I and II. Additionally, the EOIMF generated by Mode II is stronger; its magnitude is approximately three times larger than that of Mode I.

3.3. EOIMF sensitivity to ice and ocean thicknesses

Here, we focus on the sensitivity of EOIMF to the ocean and ice thicknesses. Fig. 8 shows the EOIMF surface amplitude calculated for the thickness models A–D (Table 1) using the default mantle, ice, and ocean conductivities from the previous section. The convection is assumed to be in Mode II.

As we can see, the EOIMF is stronger if the ocean is thicker and the ice is thinner. The reason is that the ocean flow is more vigorous in the thicker ocean to balance the decrease in the conductive heat flux. In addition, the thicker ice damps the field by shifting the source away from the surface (see Section 3.4).

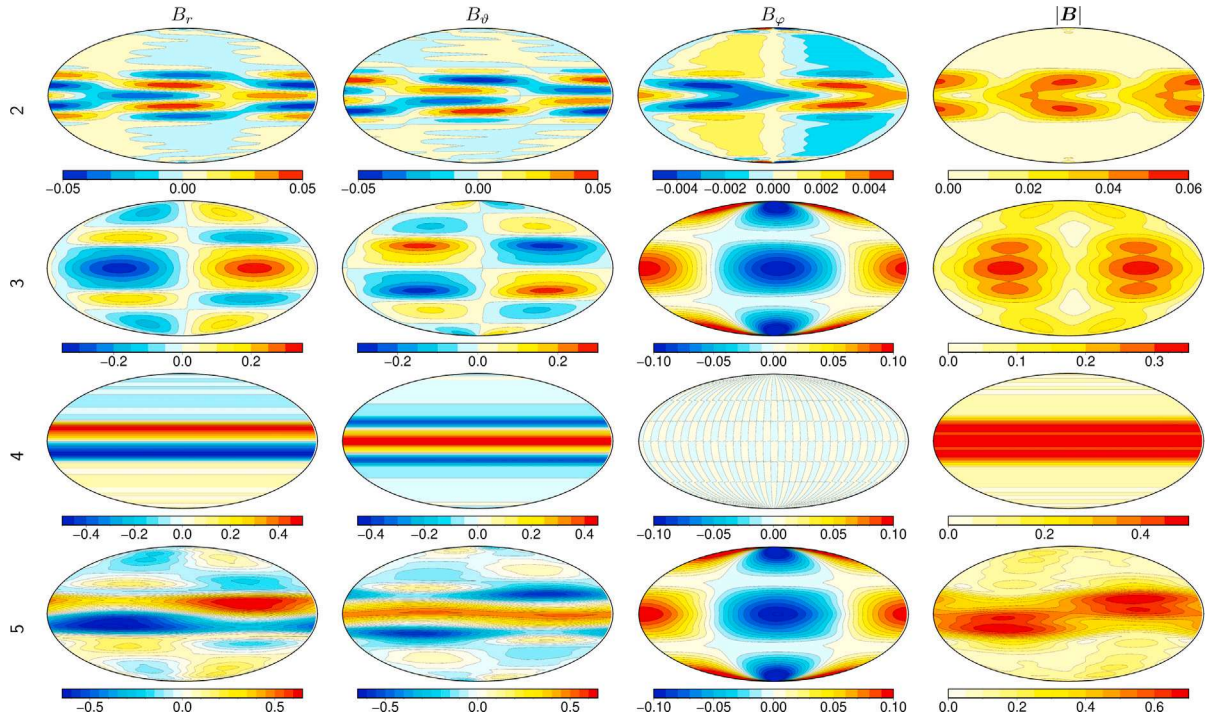


Fig. 5. EOIMF [nT] for Configurations 2–5 (Table 2) calculated using thickness model A and flow Mode II in time $t = t_{spin} + 4$ h.

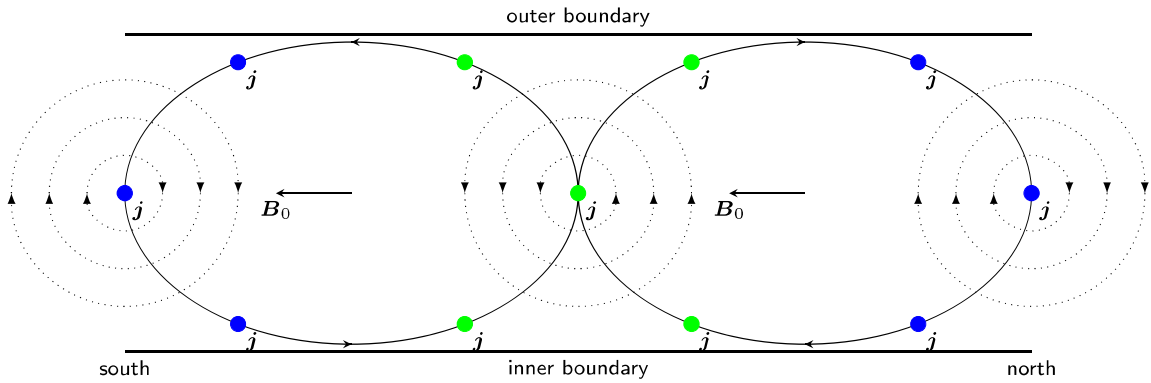


Fig. 6. Sketch of the convection cells (solid ellipses), the electrical currents (color circles; green = out of the paper, blue = into the paper), and corresponding OIMF (dotted circles) in a meridional plane. The curvature of ocean boundaries is neglected for simplicity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

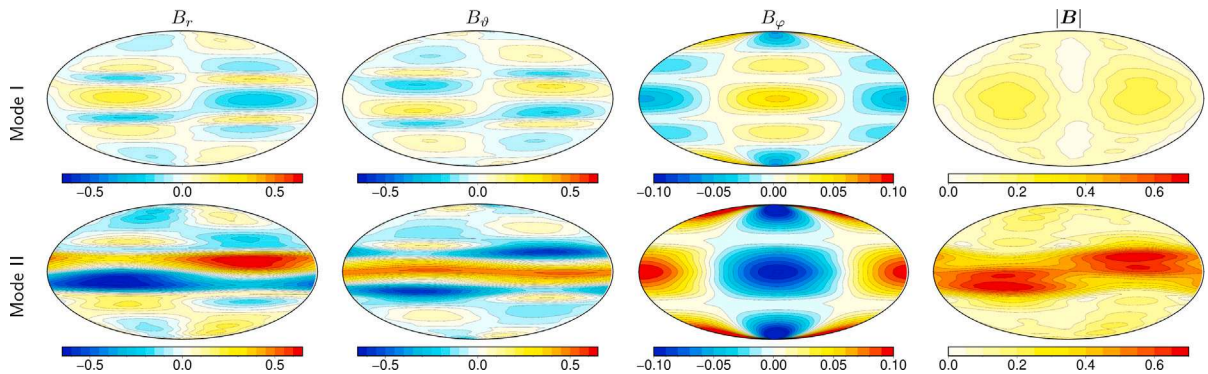


Fig. 7. EOIMFs [nT] calculated using thickness model A and one of two possible flow configurations shown in Fig. 1 in time $t = t_{spin} + 4$ h.

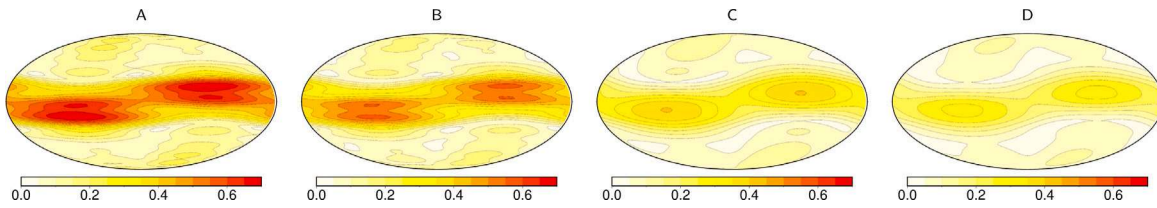


Fig. 8. EOIMF strength [nT] for thicknesses A–D from Table 1 and flow Mode II. Snapshots in time $t = t_{spin} + 4$ h.

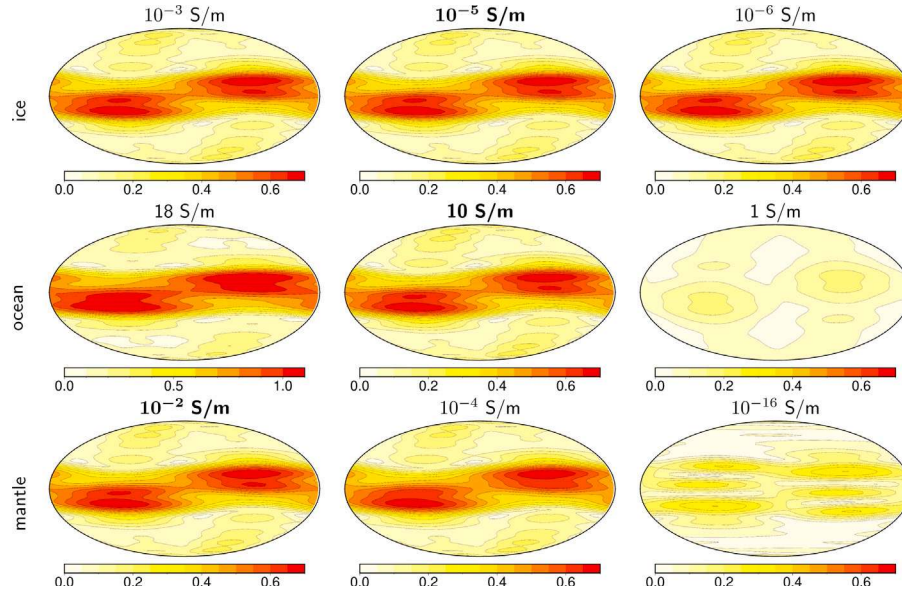


Fig. 9. EOIMF strength [nT] as a function of mantle, ocean, and ice conductivity for thickness model A. The default conductivity values are in bold. Snapshots in time $t = t_{spin} + 4$ h.

3.4. EOIMF sensitivity to mantle, ocean and ice conductivity

Here, we explore the dependence of EOIMF on the electrical conductivity of ice, ocean, and mantle. Fig. 9 summarizes the EOIMF obtained by a series of calculations in which one conductivity value was perturbed from the default model. The thicknesses are fixed to model A, and the convective flow is in Mode II.

Mantle conductivity has a minimal effect on the EOIMF (Fig. 9, mantle). If we decrease mantle conductivity by two orders of magnitude to $\sigma_m = 10^{-4}$ S/m, the EOIMF remains almost unchanged. A further decrease of mantle conductivity to $\sigma_m = 10^{-6}$ S/m results in a mild weakening of the EOIMF, around 10% in maximum values. In the case of a resistive mantle, $\sigma_m = 10^{-16}$ S/m, the EOIMF strength decreases by approximately 50%. However, this conductivity value is most likely too low.

Ocean conductivity has the expected effect on the EOIMF; a more conductive ocean implies a stronger EOIMF (Fig. 9, ocean). Indeed, within the ocean layer, the second term on the left-hand side of the EMI equation (8) can be written as $-\sigma^{-1} \Delta \mathbf{B}$. In other words, it is a diffusion term inversely proportional to conductivity. The EOIMF strength reaches 1 nT if $\sigma_o = 18$ S/m; the color scale is extended in the corresponding panel. However, the value of 18 S/m corresponds to an extremely conductive ocean, which is probably unrealistic.

Finally, ice conductivity has a negligible effect on the EOIMF in model A (Fig. 9, ice). However, this is not unexpected, given that the ice layer is only 1 km thick, and thus, its impact is minimal. Therefore, we repeat the numerical experiment for model C, where the ice thickness is 30 km (Fig. 10). Ice conductivity matters in model C. The EOIMF is stronger if ice is more conductive. Nevertheless, even if we consider a conductivity of $\sigma_i = 10^{-3}$ S/m, which is probably an exaggerated value,

the EOIMF in model C remains weaker than in model A. It is worth noting that, due to change in the ocean's thickness (D), the mean speed in model C is 0.25 m/s, compared to 0.28 m/s in model A (see Table 1), which partially explains the lower value of the magnetic field.

4. Conclusions

We have presented the first numerical calculations of Europa's ocean-induced magnetic field (EOIMF) generated by the convective flow in Europa's subsurface ocean in the presence of Jupiter's magnetic field. Our calculations indicate that even for the most favorable ocean and ice conductivities and thicknesses, the magnitude of EOIMF is below 1 nT on the Europa's surface, which is more than one order of magnitude less than estimated by Vance et al. (2021). The discrepancy is primarily attributable to the complete solution of the electromagnetic induction equation in our study (versus a simple scaling relation in Vance et al. (2021)) and the more sluggish ocean flow used to compute the EOIMF, which is a consequence of a newly derived scaling relation for the velocity field. Furthermore, we have demonstrated that the convection cells in the meridional planes may generate equally strong EOIMF as the zonal jets despite the order of magnitude smaller velocities in the convection cells.

While the ocean flow averaged over time is symmetric about the equator, only the zonal component of the calculated EOIMF is symmetric, but the radial and meridional components show no symmetry. Nevertheless, partial EOIMFs generated by the vertical/horizontal JMF and the zonal/meridional-plane flow are symmetric or antisymmetric. In the presence of vertical JMF, the flow in convection cells generates rotationally symmetric EOIMF, which reflects (to a certain degree) the geometry of the flow: meridional velocities in the radial component and

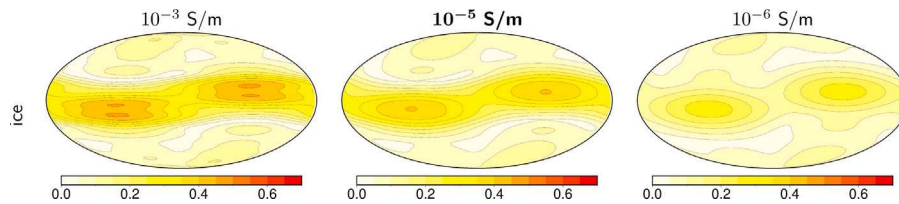


Fig. 10. EOIMF strength [nT] as a function of ice conductivity for thickness model C. The default conductivity value is in bold. Snapshots in time $t = t_{spin} + 4$ h.

vice versa. Consequently, the structures in the radial and meridional components of the total EOIMF may resemble zonal stripes.

These results have significant consequences for the Juice and Europa Clipper missions. If we successfully retrieved the EOIMF signal from the satellite measurements of the magnetic field, we could use it to address additional Level-1 scientific questions (Roberts et al., 2023; Van Hoolst et al., 2024), such as identifying the ocean flow mode from the EOIMF structure. However, our results suggest that the EOIMF signal is on the edge of detectability of the upcoming satellite missions. For example, the Europa Clipper mission has a required sensitivity of magnetic field measurements of 1.5 nT (Kivelson et al., 2023). The EOIMF signal will be further attenuated with the increasing distance from Europa's surface. For a typical Juice flyby at $h = 400$ km, the respective attenuation factors $(a/(a+h))^{j+2}$ for the SH degrees $j = 2, \dots, 6$ are 0.40, 0.32, 0.25, 0.20 and 0.16. The Europa Clipper mission is expected to get as close as 25 km, implying the attenuation factors 0.94, 0.92, 0.91, 0.89 and 0.88 for the same SH degrees. Last but not least, isolating the EOIMF from the aggregate magnetic imprint of the complex Europa's interactions with Jupiter's magnetosphere will represent a challenging problem of its own.

CRedit authorship contribution statement

L. Šachl: Writing – original draft, Investigation. J. Kvorka: Writing – original draft, Investigation. O. Čadek: Writing – original draft, Supervision. J. Velímský: Writing – original draft, Supervision.

Use of AI

During the preparation of this work the author(s) used Grammarly and DeepL in order to improve grammar and readability

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank Alexander Grayver and one anonymous reviewer for their helpful comments. This research was supported by the European Space Agency (ESA), contract No. 4000141625/23/NL/MGu/my, and the Ministry of Education, Youth, and Sports of the Czech Republic through the e-INFRA CZ (ID:90254). J.K. acknowledges the support from the Charles University project SVV, Czech Republic 260581. O.Č. is a member of the Nečas Center for Mathematical Modeling. J.V. acknowledges the funding by European Space Agency (ESA) Contract No. 4000143627/24/I-EB Swarm for Ocean Dynamics, part of the ESA Solid Earth Magnetic Science Cluster, EO for Society programme.

Appendix A. Ocean model

A.1. Determining the Rayleigh number

In the following, we assume that any quantity X characterizing the ocean model can be expressed in the form (e.g. Christensen and Aubert, 2006)

$$X \sim Pr^a Ek^b Ra_Q^c, \quad (\text{A.1})$$

where a, b, c are scaling exponents and Ra_Q is the flux-based Rayleigh number,

$$Ra_Q = \frac{1}{4\pi r_o(r_o - D)} \frac{\alpha g_o Q_{adv}}{\rho_w C_p \omega^3 D^2} = Ra(Nu - 1) Ek^3 Pr^{-2}. \quad (\text{A.2})$$

Here, r_o is the ocean outer boundary radius, Q_{adv} is the advected heat flow, ρ_w is the water density, C_p is the thermal capacity of water at constant pressure and Nu is the Nusselt number,

$$Nu = \frac{1}{4\pi r_o(r_o - D)} \frac{QD}{\rho_w C_p \kappa \Delta T}, \quad (\text{A.3})$$

where Q is the total heat flow through the ocean, containing the advected and the convective part.

Since the Rayleigh number cannot be determined directly from Eq. (4) in the main text because of the lack of information on ΔT , we derive a scaling law for Ra in the form of Eq. (A.1) using the least squares method applied to the dataset in Table A.3 (see also Fig. A.1). In Table A.1, we compare the scaling law derived in this study with those used in the literature and present the corresponding ranges of Ra , Ro_{loc} and Ro_c evaluated based on the parameter values detailed in the main text. The estimated Rayleigh numbers span approximately two orders of magnitude, $9.6 \cdot 10^{19} - 3.9 \cdot 10^{22}$, which is comparable to the range of $10^{20} - 10^{22}$ derived by Soderlund (2019), where the lower bound corresponds to the scaling law for the rapidly rotating convection (TG16) while the upper bound is obtained from the scaling law for a non-rotating ocean (TG15). The values in Table A.1 indicate that Europa's ocean is in the transitional regime, as $Ro_{loc} \gg 1$ and $Ro_c \ll 1$.

Therefore, the scaling laws TG15 and TG16 are not applicable because they are relevant to non-rotating and rapidly rotating regimes, respectively. The scaling laws CH02 and KC22 may only hold within a limited range near the onset of convection (Cheng and Aurnou, 2016) and, thus, may be inappropriate for Europa. The scaling law LS24 used in this study bridges the gap between the non-diffusive scaling laws suitable for rapidly rotating oceans on the one side and non-rotating oceans on the other, reducing the range of admissible Ra by an order of magnitude. LS24 is suitable for subsurface oceans in the transitional regime, such as Europa, unlike TG15 and TG16, which have been derived for different convection regimes.

A.2. Velocity of ocean flow

The velocity of the ocean flow can be estimated using Eq. (A.1) with $X = Ro$, where Ro is the Rossby number, defined as the root mean square velocity of the ocean flow, U , scaled by ωD

$$Ro = \frac{U}{\omega D}. \quad (\text{A.4})$$

Table A.1

Selected scaling laws proposed for estimating the Rayleigh number (column 3) and the values of Ra , Ro_{loc} , and Ro_c corresponding to Europa's ocean (columns 4–6). For the definitions of Ro_{loc} and Ro_c , see Eqs. (5) and (6) in the main text. The original scaling laws presented in the cited literature have been rewritten to the form of Eq. (A.1), where we assumed $Nu \gg 1$ in Eq. (A.2). The scaling laws CH02 and KC22 were derived for models with free-slip boundaries without assuming a priori a specific convection regime. In contrast, TG15 and TG16 were derived for non-rotating and rapidly rotating no-slip models, respectively. LS24 is derived from simulations with free-slip boundaries placed in the transitional regime. The presented scaling law also differs in the value of Pr for which they are applicable: CH02, KC22, and LS24 were derived considering a range of Pr , whereas TG15 and TG16 assumed $Pr=1$. The dependence on Pr in TG15 and TG16 results from using a molecular value, $Pr=10$, in Eq. (A.2), a practice frequently used in the literature (e.g., Soderlund et al., 2014; Soderlund, 2019; Amit et al., 2020; Cabanes et al., 2024).

Notation, reference	Radius ratio	Scaling law	Ra	Ro_{loc}	Ro_c
CH02, Christensen (2002)	0.35	$Ra = 12.99 Ek^{-2.00} Pr^{1.00} Ra_Q^{0.45}$	$3.7 \cdot 10^{20} - 4.7 \cdot 10^{21}$	203–607	0.05–0.08
TG15, Gastine et al. (2015)	0.60	$Ra = 7.35 Ek^{-2.25} Pr^{1.5} Ra_Q^{0.75}$	$1.5 \cdot 10^{21} - 3.9 \cdot 10^{22}$	827–5068	0.11–0.23
TG16, Gastine et al. (2016)	0.60	$Ra = 2.14 Ek^{-2.00} Pr^{0.8} Ra_Q^{0.40}$	$9.6 \cdot 10^{19} - 1.1 \cdot 10^{21}$	52–144	0.03–0.04
KC22, Kverka and Čadež (2022)	0.80	$Ra = 14.73 Ek^{-2.00} Pr^{1.0} Ra_Q^{0.53}$	$9.6 \cdot 10^{19} - 1.4 \cdot 10^{21}$	52–177	0.03–0.04
LS24, this study	0.90	$Ra = 4.61 Ek^{-2.19} Pr^{1.27} Ra_Q^{0.62}$	$1.3 \cdot 10^{21} - 2.7 \cdot 10^{22}$	728–3416	0.10–0.18

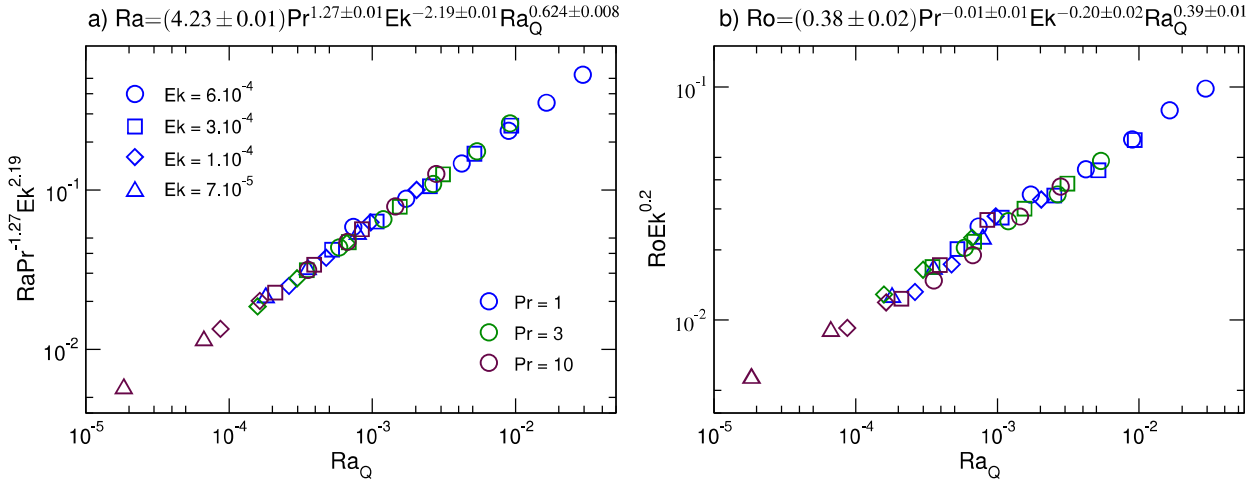


Fig. A.1. Graphs illustrating the scaling laws derived for (a) Ra and (b) Ro from the dataset in Table A.3 using the least squares method. Since Ro depends only weakly on the Prandtl number, the dependence on Pr is omitted. Shapes and colors of the symbols indicate the values of parameters Ek and Pr used in the numerical simulation.

The scaling law for Ro is again obtained by the least squares analysis of the dataset in Table A.3 ($Ro = 0.38 Ek^{-2} Ra_Q^{0.39}$, see Fig. A.1b). In order to determine U , we first calculate Ro from the above scaling law into which we substitute the values of Ra_Q , Eq. (A.2), and Ek , Eq. (4), corresponding to Europa's ocean, and then calculate U from Eq. (A.4). Values for U obtained in this way range from 8 to 28 cm/s. This range is consistent with the findings of Lemasquerier et al. (2023), who reported velocities in a range of 3.5–22 cm/s for heterogeneous heat flux conditions at the bottom boundary. The analysis of energetic constraints by Jansen et al. (2023) predicts velocities of 1.5–6 cm/s, slightly lower than those obtained by LS24. Considering the eddy viscosities of $O(10^{-3})$ m²/s (Bire et al., 2022) found velocities in the range of mm/s. In contrast to other GCM studies that have implemented the eddy viscosity concept (Kang, 2022; Kang and Jansen, 2022; Ashkenazy and Tziperman, 2021; Ashkenazy et al., 2023) and assumed varying horizontal and vertical viscosities, Bire et al. (2022) assume a constant viscosity throughout the ocean volume. This allows the results of Bire et al. (2022) to be directly compared with those obtained in the present study using the DNS approach. Considering the eddy viscosity and diffusivity, as in Bire et al. (2022), we get $Pr=1$ and $Ek \approx 10^{-9} - 10^{-8}$. For these values, the LS24 scaling law predicts velocities ranging from 1 to 8 cm/s. These velocities are consistent with Jansen et al. (2023) but approximately ten times higher than those derived by Bire et al. (2022) for a freshwater ocean. For completeness, it should also be mentioned that Ashkenazy and Tziperman (2021) used the MITgcm code in a Europa setting, incorporating salinity and phase boundary effects, and reported velocities of the order of cm/s. For detailed information on the results discussed above, see Table A.2.

Appendix B. EOIMF

B.1. The Lorentz force for configurations 1–4

In this section, we provide the analytical expressions for the Lorentz force, $\mathbf{L}$, and the right-hand side ($\nabla \times \mathbf{L}$) of Eq. (8) corresponding to Configurations 1–4 defined in Table 2. In Configuration 1, the Lorentz force is

$$\mathbf{L}_1 = v_\varphi B_{0\varphi}^v \mathbf{e}_r + v_\varphi B_{0r}^v \mathbf{e}_\varphi, \quad (\text{B.1})$$

where L_{1r} is symmetric and $L_{1\vartheta}$ is antisymmetric since the product of two symmetric (or two antisymmetric) fields is symmetric and the product of one symmetric and one antisymmetric field is antisymmetric. The right-hand side $\mathbf{R}_1$ of the EMI equation is then

$$\mathbf{R}_1(r, \vartheta) = \nabla \times \mathbf{L}_1 = \frac{1}{r} \left[L_{1\vartheta} - \frac{\partial L_{1r}}{\partial \vartheta} \right] \mathbf{e}_\varphi, \quad (\text{B.2})$$

which is antisymmetric with respect to the equator since a meridional derivative of a symmetric function is antisymmetric (and vice versa).

In Configuration 2, the Lorentz force is

$$\mathbf{L}_2 = v_\vartheta B_{0\vartheta}^h \mathbf{e}_r - v_r B_{0\vartheta}^h \mathbf{e}_\vartheta + (v_r B_{0\vartheta}^h - v_\vartheta B_{0r}^h) \mathbf{e}_\varphi, \quad (\text{B.3})$$

where $L_{2\vartheta}$ is symmetric and L_{2r} and $L_{2\varphi}$ are antisymmetric. The curl of the Lorentz force is

$$\begin{aligned} \mathbf{R}_2 = \frac{1}{r \sin \vartheta} \left[\frac{\partial}{\partial \vartheta} (L_{2\varphi} \sin \vartheta) - \frac{\partial L_{2\vartheta}}{\partial \varphi} \right] \mathbf{e}_r + \frac{1}{r} \left[\frac{1}{\sin \vartheta} \frac{\partial L_{2r}}{\partial \varphi} - \frac{\partial}{\partial r} (r L_{2\varphi}) \right] \mathbf{e}_\vartheta \\ + \frac{1}{r} \left[\frac{\partial}{\partial r} (r L_{2\vartheta}) - \frac{\partial L_{2r}}{\partial \vartheta} \right] \mathbf{e}_\varphi, \end{aligned} \quad (\text{B.4})$$

where R_{2r} and $R_{2\varphi}$ are symmetric and $R_{2\vartheta}$ is antisymmetric.

Table A.2

Estimates of the mean flow speed in Europa's ocean (column 4) derived from published scaling laws, energetic constraints, and GCM simulations.

Notation, reference	Radius ratio	Method, scaling law	U
CH02, Christensen (2002)	0.35	DNS, $Ro = 0.65 Ra_Q^{0.20}$	3–7 cm/s
TG15, Gastine et al. (2015)	0.60	DNS, $Ro = 0.88 Ek^{-0.1} Pr^{0.7} Ra_Q^{0.36}$	15–46 cm/s
AS21, Ashkenazy and Tziperman (2021)	0.92	GCM	~ cm/s
BI22, Bire et al. (2022)	0.76–0.96	GCM	~ mm/s
JA23, Jansen et al. (2023)	0.92	energetic constraints	1.5–6 cm/s
LE23, Lemasquerier et al. (2023)	0.90	DNS, not reported	3.5–22 cm/s
CA24, Cabanes et al. (2024)	0.80	DNS, not reported	21–61 cm/s
LS24, this study	0.90	DNS, $Ro = 0.38 Ek^{-0.2} Ra_Q^{0.39}$	8–28 cm/s

Table A.3

List of numerical simulations used to derive the scaling laws for Europa's ocean. Symbols N_r and j_{max} denote the number of the Chebyshev points and the cut-off degree used to discretize Eqs. (1)–(3) in the radial and lateral directions, respectively. The results of the simulations are represented by the Nusselt (Nu) and Rossby (Ro) numbers.

Model No.	N_r	j_{max}	Pr	Ek	Ra	Ro_{loc}	Ro_c	R_M	Nu	Ro
1	61	157	1	$6 \cdot 10^{-4}$	$6.67 \cdot 10^5$	4.67	0.49	2.00	5.11	0.11
2	61	157	1	$6 \cdot 10^{-4}$	$1.00 \cdot 10^6$	7.00	0.60	3.00	7.95	0.15
3	73	177	1	$6 \cdot 10^{-4}$	$1.67 \cdot 10^6$	11.67	0.77	5.00	11.67	0.19
4	73	177	1	$6 \cdot 10^{-4}$	$2.67 \cdot 10^6$	18.68	0.98	8.00	15.49	0.26
5	73	207	1	$6 \cdot 10^{-4}$	$4.00 \cdot 10^6$	28.02	1.20	12.00	18.95	0.35
6	85	257	1	$6 \cdot 10^{-4}$	$6.00 \cdot 10^6$	42.02	1.47	18.00	22.68	0.43
7	73	177	1	$3 \cdot 10^{-4}$	$2.19 \cdot 10^6$	5.05	0.44	2.00	8.84	0.10
8	73	177	1	$3 \cdot 10^{-4}$	$3.28 \cdot 10^6$	7.58	0.54	3.00	12.09	0.14
9	73	207	1	$3 \cdot 10^{-4}$	$5.47 \cdot 10^6$	12.64	0.70	5.00	17.05	0.17
10	81	237	1	$3 \cdot 10^{-4}$	$8.76 \cdot 10^6$	20.22	0.89	8.00	21.78	0.22
11	81	285	1	$3 \cdot 10^{-4}$	$1.31 \cdot 10^7$	30.32	1.09	12.00	26.29	0.30
12	73	217	1	$1 \cdot 10^{-4}$	$1.44 \cdot 10^7$	5.73	0.38	2.00	18.17	0.08
13	73	247	1	$1 \cdot 10^{-4}$	$2.16 \cdot 10^7$	8.60	0.46	3.00	22.01	0.11
14	73	377	1	$1 \cdot 10^{-4}$	$3.60 \cdot 10^7$	14.33	0.60	5.00	27.00	0.18
15	97	413	1	$1 \cdot 10^{-4}$	$5.76 \cdot 10^7$	22.92	0.76	8.00	35.32	0.21
16	81	257	1	$7 \cdot 10^{-5}$	$2.65 \cdot 10^7$	5.97	0.36	2.00	19.80	0.08
17	81	257	1	$7 \cdot 10^{-5}$	$3.98 \cdot 10^7$	8.95	0.44	3.00	25.98	0.11
18	73	347	1	$7 \cdot 10^{-5}$	$6.63 \cdot 10^7$	14.92	0.57	5.00	34.69	0.15
19	65	177	3	$6 \cdot 10^{-4}$	$2.00 \cdot 10^6$	7.25	0.49	2.63	12.23	0.09
20	65	177	3	$6 \cdot 10^{-4}$	$3.00 \cdot 10^6$	10.87	0.60	3.95	16.54	0.12
21	73	197	3	$6 \cdot 10^{-4}$	$5.00 \cdot 10^6$	18.11	0.77	6.58	22.13	0.15
22	73	197	3	$6 \cdot 10^{-4}$	$8.01 \cdot 10^6$	28.98	0.98	10.53	27.96	0.21
23	73	227	3	$3 \cdot 10^{-4}$	$6.57 \cdot 10^6$	7.84	0.44	2.63	17.69	0.09
24	73	227	3	$3 \cdot 10^{-4}$	$9.85 \cdot 10^6$	11.76	0.54	3.95	23.17	0.11
25	73	227	3	$3 \cdot 10^{-4}$	$1.64 \cdot 10^7$	19.61	0.70	6.58	31.55	0.15
26	85	267	3	$3 \cdot 10^{-4}$	$2.63 \cdot 10^7$	31.37	0.89	10.53	39.57	0.20
27	85	267	3	$1 \cdot 10^{-4}$	$4.32 \cdot 10^7$	8.89	0.38	2.63	32.96	0.08
28	85	267	3	$1 \cdot 10^{-4}$	$6.48 \cdot 10^7$	13.34	0.46	3.95	41.45	0.10
29	85	297	3	$1 \cdot 10^{-4}$	$1.08 \cdot 10^8$	22.23	0.60	6.58	55.25	0.14
30	73	177	10	$6 \cdot 10^{-4}$	$6.67 \cdot 10^6$	11.73	0.49	3.56	24.68	0.07
31	85	237	10	$6 \cdot 10^{-4}$	$1.00 \cdot 10^7$	17.59	0.60	5.33	30.99	0.08
32	85	237	10	$6 \cdot 10^{-4}$	$1.67 \cdot 10^7$	29.32	0.77	8.89	40.08	0.12
33	97	267	10	$6 \cdot 10^{-4}$	$2.67 \cdot 10^7$	46.91	0.98	14.23	48.36	0.16
34	85	237	10	$3 \cdot 10^{-4}$	$2.19 \cdot 10^7$	12.70	0.44	3.56	35.50	0.06
35	85	237	10	$3 \cdot 10^{-4}$	$3.28 \cdot 10^7$	19.04	0.54	5.33	44.27	0.09
36	97	285	10	$3 \cdot 10^{-4}$	$5.47 \cdot 10^7$	31.74	0.70	8.89	57.30	0.14
37	97	267	10	$1 \cdot 10^{-4}$	$1.44 \cdot 10^8$	14.39	0.38	3.56	60.53	0.06
38	97	267	10	$1 \cdot 10^{-4}$	$2.16 \cdot 10^8$	21.59	0.46	5.33	75.77	0.08
39	97	285	10	$7 \cdot 10^{-5}$	$1.33 \cdot 10^8$	7.50	0.25	1.78	40.45	0.04
40	113	321	10	$7 \cdot 10^{-5}$	$2.65 \cdot 10^8$	14.99	0.36	3.56	73.29	0.06

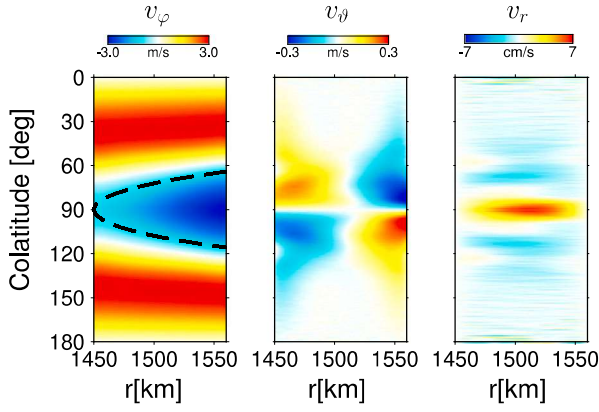


Fig. B.1. Flow model Sod19.

In Configuration 3, the Lorentz force is

$$\mathbf{L}_3 = v_\phi B_{0\theta}^h \mathbf{e}_r + v_\phi B_{0r}^h \mathbf{e}_\phi, \quad (\text{B.5})$$

where $L_{3\theta}$ is symmetric and L_{3r} is antisymmetric. The curl of the Lorentz force is

$$\mathbf{R}_3 = -\frac{1}{r \sin \theta} \frac{\partial L_{3\theta}}{\partial \phi} \mathbf{e}_r + \frac{1}{r \sin \theta} \frac{\partial L_{3r}}{\partial \phi} \mathbf{e}_\theta + \frac{1}{r} \left(L_{3\theta} - \frac{\partial L_{3r}}{\partial \theta} \right) \mathbf{e}_\phi, \quad (\text{B.6})$$

where R_{3r} and $R_{3\phi}$ are symmetric and $R_{3\theta}$ is antisymmetric.

In Configuration 4, the Lorentz force is symmetric,

$$\mathbf{L}_4 = (v_r B_{0\theta}^v - v_\theta B_{0r}^v) \mathbf{e}_\phi, \quad (\text{B.7})$$

and its curl is

$$\mathbf{R}_4 = \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (L_{4\phi} \sin \theta) \mathbf{e}_r - \frac{1}{r} \frac{\partial}{\partial r} (r L_{4\phi}) \mathbf{e}_\theta, \quad (\text{B.8})$$

where $R_{4\theta}$ is symmetric and R_{4r} is antisymmetric.

B.2. Comparison with the model of Vance et al. (2021)

In this section, we compare our EOIMF results with the estimate provided by Vance et al. (2021). For this purpose, we consider the same flow model as in Vance et al. (2021), hereinafter referred to as Sod19, and use it to calculate the corresponding EOIMF by solving the EMI equation (see Section 2.2). The flow model was first presented by Soderlund (2019) and we obtained it by solving Eqs. (1)–(3) for $Ek = 3 \cdot 10^{-4}$, $Ra = 3.4 \cdot 10^7$, $Pr = 1$ and $r_i/r_o = 0.9$ using our numerical code. Comparison of Sod19 (Fig. B.1) with the model used in the present study (Fig. 1b) shows that both models are characterized by a strong retrograde zonal flow outside the tangential cylinder, a pair of prograde jets at higher latitude, and a non-negligible circulation in the meridional plane. However, the models differ in the flow speed, which is about ten times higher for Sod19 than for the model used in this study.

The EOIMF obtained for Sod19 by solving the EMI equation for the thickness model A is presented in Fig. B.2. The figure is organized in the same way as Fig. 5 in the main text, which allows a detailed comparison of both fields. While the differences in EOIMF patterns are minor, the magnitude of EOIMF obtained for Sod19 ($\lesssim 7$ nT) is almost an order of magnitude larger than that predicted for our flow model and a factor of three smaller than that reported by Vance et al. (2021) (20 nT, see Table 5 therein). The former discrepancy is related to the fact that Vance et al. (2021) used Sod19 without scaling the flow speed, whereas the speed in our flow model is corrected using the appropriate scaling law (see Appendix A.2). The latter discrepancy can be attributed to the differences in the methods used to calculate the motionally induced field strength. The estimate of Vance et al. (2021) is calculated using the formula $B \sim \mu_0 \sigma D U B_0$, unlike our EOIMF estimate, which is based on the solution of the general EMI equation.

Data availability

Modeling results can be downloaded from <https://zenodo.org/records/13838989>.

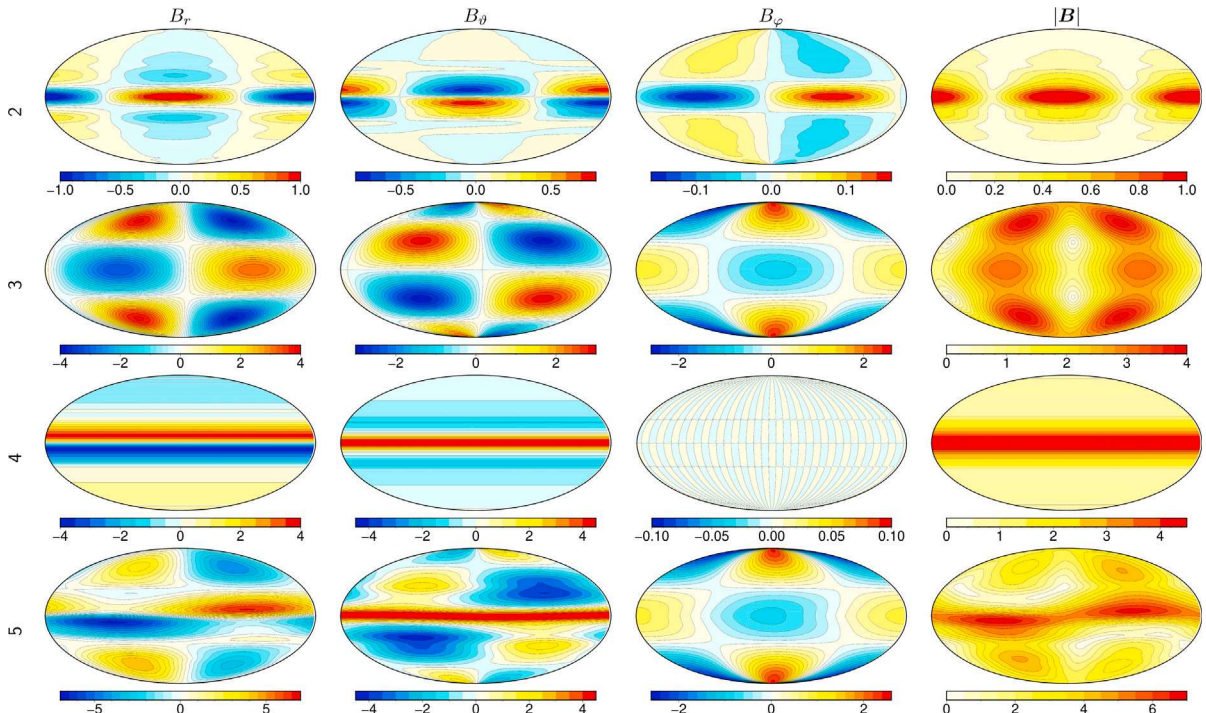


Fig. B.2. EOIMF [nT] four hours after the spin up calculated for the flow model Sod19. The figure is organized in the same way as Fig. 5 in the main text.

References

- Amit, H., Choblet, G., Tobie, G., Terra-Nova, F., Čadek, O., Bouffard, M., 2020. Cooling patterns in rotating thin spherical shells – Application to Titan's subsurface ocean. *Icarus* 338, 113509. <http://dx.doi.org/10.1016/j.icarus.2019.113509>.
- Ashkenazy, Y., Tziperman, E., Nimmo, F., 2023. Non-synchronous rotation on Europa driven by ocean currents. *AGU Adv.* 4, e2022AV000849. <http://dx.doi.org/10.1029/2022AV000849>.
- Aurnou, J., Heimpel, M., Wicht, J., 2007. The effects of vigorous mixing in a convective model of zonal flow on the ice giants. *Icarus* 190, 110–126. <http://dx.doi.org/10.1016/j.icarus.2007.02.024>.
- Běhouňková, M., Tobie, G., Choblet, G., Kervazo, M., Melwani Daswani, M., Dumoulin, C., Vance, S.D., 2021. Tidally induced magmatic pulses on the oceanic floor of Jupiter's moon Europa. *Geophys. Res. Lett.* 48, e2020GL090077. <http://dx.doi.org/10.1029/2020GL090077>.
- Billings, S.E., Kattenhorn, S.A., 2005. The great thickness debate: Ice shell thickness models for Europa and comparisons with estimates based on flexure at ridges. *Icarus* 177 (2), 397–412. <http://dx.doi.org/10.1016/j.icarus.2005.03.013>.
- Bire, S., Kang, W., Ramadhan, A., Campin, J.-M., Marshall, J., 2022. Exploring ocean circulation on icy moons heated from below. *J. Geophys. Res.: Planets* 127, e2021JE007025. <http://dx.doi.org/10.1029/2021JE007025>.
- Bryan, K., 1969. A numerical method for the study of the circulation of the world ocean. *J. Comput. Phys.* 4, 347–376. [http://dx.doi.org/10.1016/0021-9991\(69\)90004-7](http://dx.doi.org/10.1016/0021-9991(69)90004-7).
- Cabanes, S., Gastine, T., Fournier, A., 2024. Zonostrophic turbulence in the subsurface oceans of the Jovian and Saturnian moons. *Icarus* 415, 116047. <http://dx.doi.org/10.1016/j.icarus.2024.116047>.
- Casajus, L.G., Zannoni, M., Modenini, D., Tortora, P., Nimmo, F., Van Hoolst, T., Buccino, D., Oudrhiri, K., 2021. Updated Europa gravity field and interior structure from a reanalysis of Galileo tracking data. *Icarus* 358, 114187. <http://dx.doi.org/10.1016/j.icarus.2020.114187>.
- Castillo-Rogez, J.C., Daswani, M.M., Glein, C.R., Vance, S.D., Cochrane, C.J., 2022. Contribution of non-water ices to salinity and electrical conductivity in ocean worlds. *Geophys. Res. Lett.* 49 (16), e2021GL097256. <http://dx.doi.org/10.1029/2021GL097256>.
- Cheng, J.S., Aurnou, J.M., 2016. Tests of diffusion-free scaling behaviors in numerical dynamo datasets. *Earth Planet. Sci. Lett.* 436, 121–129. <http://dx.doi.org/10.1016/j.epsl.2015.12.004>.
- Cheng, J.S., Aurnou, J.M., Julien, K., Kunnen, R.P.J., 2018. A heuristic framework for next-generation models of geostrophic convective turbulence. *Geophys. Astrophys. Fluid Dyn.* 112, 277–300. <http://dx.doi.org/10.1080/03091929.2018.1506024>.
- Christensen, U.R., 2002. Zonal flow driven by strongly supercritical convection in rotating spherical shells. *J. Fluid Mech.* 470, 115–133. <http://dx.doi.org/10.1017/S0022112002002008>.
- Christensen, U.R., Aubert, J., 2006. Scaling properties of convection-driven dynamos in rotating spherical shells and application to planetary magnetic fields. *Geophys. J. Int.* 166, 97–114. <http://dx.doi.org/10.1111/j.1365-246X.2006.03009.x>.
- Christensen, U.R., Wicht, J., 2007. 8.08 - numerical dynamo simulations. In: Schubert, Gerald (Ed.), *Treatise on Geophysics*. Elsevier, Amsterdam, ISBN: 978-0-444-52748-6, pp. 245–282. <http://dx.doi.org/10.1016/B978-0-444-52748-6.00134-6>.
- Colburn, D.S., Reynolds, R.T., 1985. Electrolytic currents in Europa. *Icarus* 63 (1), 39–44. [http://dx.doi.org/10.1016/0019-1035\(85\)90019-3](http://dx.doi.org/10.1016/0019-1035(85)90019-3).
- Connerney, J.E.P., Timmins, S., Oliverson, R.J., Espley, J.R., Joergensen, J.L., Kotisarios, S., Joergensen, P.S., Merayo, J.M.G., Herceg, M., Bloxham, J., Moore, K.M., Mura, A., Moirano, A., Bolton, S.J., Levin, S.M., 2022. A new model of Jupiter's magnetic field at the completion of Juno's Prime Mission. *J. Geophys. Res.: Planets* 127 (2), e2021JE007055. <http://dx.doi.org/10.1029/2021JE007055>.
- Cox, M.D., 1984. *A Primitive Equation, 3-Dimensional Model of the Ocean*. GFDL Ocean Group Technical Report No. 1, Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey.
- Cox, R.A., McCartney, M.J., Culkin, F., 1970. The specific gravity/salinity/temperature relationship in natural seawater. *Deep-Sea Res.* 17, 679–689. [http://dx.doi.org/10.1016/0011-7471\(70\)90034-3](http://dx.doi.org/10.1016/0011-7471(70)90034-3).
- Flosadóttir, Á.H., Larsen, J.C., Smith, J.T., 1997. Motional induction in North Atlantic circulation models. *J. Geophys. Res.: Ocean* 102 (C5), 10353–10372. <http://dx.doi.org/10.1029/96JC03603>.
- Gastine, T., Wicht, J., Aubert, J., 2016. Scaling regimes in spherical shell rotating convection. *J. Fluid Mech.* 808, 690–732. <http://dx.doi.org/10.1017/jfm.2016.659>.
- Gastine, T., Wicht, J., Aurnou, J.M., 2015. Turbulent Rayleigh–Bénard convection in spherical shells. *J. Fluid Mech.* 778, 721–764. <http://dx.doi.org/10.1017/jfm.2015.401>.
- Gilman, P.A., 1977. Nonlinear Dynamics of Boussinesq Convection in a Deep Rotating Spherical Shell-I. *Geophys. Astrophys. Fluid Dyn.* 8, 93–135. <http://dx.doi.org/10.1080/03091927708240373>.
- Goodman, J.C., Collins, G.C., Marshall, J., Pierrehumbert, R.T., 2004. Hydrothermal plume dynamics on Europa: Implications for chaos formation. *J. Geophys. Res.: Planets* 109, E03008. <http://dx.doi.org/10.1029/2003JE002073>.
- Goodman, J.C., Lenferink, E., 2012. Numerical simulations of marine hydrothermal plumes for Europa and other icy worlds. *Icarus* 221, 970–983. <http://dx.doi.org/10.1016/j.icarus.2012.08.027>.
- Grayver, A.V., Munch, F.D., Kuvshinov, A.V., Khan, A., Sabaka, T.J., Tøffner-Clausen, L., 2017. Joint inversion of satellite-detected tidal and magnetospheric signals constrains electrical conductivity and water content of the upper mantle and transition zone. *Geophys. Res. Lett.* 44, 6074–6081. <http://dx.doi.org/10.1002/2017GL073446>.
- Grayver, A.V., Olsen, N., 2019. The magnetic signatures of the M₂, N₂, and O₁ oceanic tides observed in Swarm and CHAMP satellite magnetic data. *Geophys. Res. Lett.* 46 (8), 4230–4238. <http://dx.doi.org/10.1029/2019GL082400>.
- Grayver, A.V., Schnepf, N.R., Kuvshinov, A.V., Sabaka, T.J., Manoj, C., Olsen, N., 2016. Satellite tidal magnetic signals constrain oceanic lithosphere-asthenosphere boundary. *Sci. Adv.* 2, e1600798. <http://dx.doi.org/10.1126/sciadv.1600798>.
- Griffies, S.M., 2004. *Fundamentals of Ocean Climate Models*. Princeton University Press, Princeton. <http://dx.doi.org/10.2307/j.ctv301gzg>.
- Hand, K.P., Chyba, C.F., 2007. Empirical constraints on the salinity of the European ocean and implications for a thin ice shell. *Icarus* 189 (2), 424–438. <http://dx.doi.org/10.1016/j.icarus.2007.02.002>.
- Howell, S.M., 2021. The likely thickness of Europa's icy shell. *Planet. Sci. J.* 2 (4), 129. <http://dx.doi.org/10.5194/epsc2020-173>.
- Jansen, M.F., Kang, W., Kite, E.S., Zeng, Y., 2023. Energetic constraints on ocean circulations of icy ocean worlds. *Planet. Sci. J.* 4, 117. <http://dx.doi.org/10.3847/PJ.2023.04.0117>.
- Julien, K., Knobloch, E., Rubio, A.M., Vasil, G.M., 2012. Heat transport in low-Rossby-number Rayleigh–Bénard convection. *Phys. Rev. Lett.* 109, 254503. <http://dx.doi.org/10.1103/PhysRevLett.109.254503>.
- Kang, W., 2022. Different ice-shell geometries on Europa and Enceladus due to their different sizes: Impacts of ocean heat transport. *Astrophys. J.* 934, 116. <http://dx.doi.org/10.3847/1538-4357/ac779c>.
- Kang, W., 2024. Nonsynchronous rotation of icy moon ice shells: The thermal wind perspective. *Sci. Adv.* 10, eadk2277. <http://dx.doi.org/10.1126/sciadv.adk2277>.
- Kang, W., Jansen, M., 2022. On icy ocean worlds, size controls ice shell geometry. *Astrophys. J.* 935, 103. <http://dx.doi.org/10.3847/1538-4357/ac7a32>.
- Khurana, K.K., Kivelson, M.G., Hand, K.P., Russell, C.T., 2009. Electromagnetic induction from Europa's ocean and the deep interior. In: Pappalardo, R.T., McKinnon, W.B., Khurana, K.K. (Eds.), *Europa*. University of Arizona Press Tucson, AZ, Tucson, pp. 572–586. <http://dx.doi.org/10.2307/j.ctt1xp3wdw.30>.
- Khurana, K.K., Kivelson, M.G., Russell, C.T., 2002. Searching for liquid water in Europa by using surface observatories. *Astrobiology* 2 (1), 93–103. <http://dx.doi.org/10.1089/153110702753621376>.
- Khurana, K.K., Kivelson, M.G., Stevenson, D.J., Schubert, G., Russell, C.T., Walker, R.J., Polansky, C., 1998. Induced magnetic fields as evidence for subsurface oceans in Europa and Callisto. *Nature* 395, 777–780. <http://dx.doi.org/10.1038/27394>.
- Kimura, J., 2024. Europa's structural conditions for the existence of subsurface ocean and the absence of metallic core-driven magnetic field. *Planet. Space Sci.* 243, 105868. <http://dx.doi.org/10.1016/j.pss.2024.105868>.
- Kivelson, M.G., Jia, X., Lee, K.A., Raymond, C.A., Khurana, K.K., Perley, M.O., Biersteker, J.B., Blackberg, J., Caron, R., Cochrane, C.J., Dawson, C.D.K., Jones, J.E., Joy, S., Korth, H., Liu, J., Maghsoudi, E., Murphy, D., Pierce, D.R., Rachos, C., Richter, I., Russell, C.T., Sherman, S., Strangeway, R.J., Villarreal, M., Weiss, B.P., Wigglesworth, L., 2023. The Europa Clipper Magnetometer. *Space Sci. Rev.* 219 (6), 48. <http://dx.doi.org/10.1007/s11214-023-00989-5>.
- Kivelson, M.G., Khurana, K.K., Joy, S., Russell, C.T., Southwood, D.J., Walker, R.J., Polansky, C., 1997. Europa's magnetic signature: Report from Galileo's pass on 19 December 1996. *Science* 276 (5316), 1239–1241. <http://dx.doi.org/10.1126/science.276.5316.1239>.
- Kivelson, M.G., Khurana, K.K., Russell, C.T., Volwerk, M., Walker, R.J., Zimmer, C., 2000. Galileo magnetometer measurements: A stronger case for a subsurface ocean at Europa. *Science* 289 (5483), 1340–1343. <http://dx.doi.org/10.1126/science.289.5483.1340>.
- Kivelson, M.G., Khurana, K.K., Volwerk, M., 2002. The permanent and inductive magnetic moments of Ganymede. *Icarus* 157 (2), 507–522. <http://dx.doi.org/10.1006/icar.2002.6834>.
- Kobe, K.A., Sheehy, T.M., 1948. Thermochemistry of sodium carbonate and its solutions. *Ind. Eng. Chem.* 40 (1), 99–102. <http://dx.doi.org/10.1021/ie50457a029>.
- Kvorka, J., Čadek, O., 2022. A numerical model of convective heat transfer in Titan's subsurface ocean. *Icarus* 376, 114853. <http://dx.doi.org/10.1016/j.icarus.2021.114853>.
- Kvorka, J., Čadek, O., 2024. The role of subsurface ocean dynamics and phase transitions in forming the topography of icy moons. *Icarus* 412, 115985. <http://dx.doi.org/10.1016/j.icarus.2024.115985>.
- Larsen, J.C., Sanford, T.B., 1985. Florida current volume transports from voltage measurements. *Science* 227 (4684), 302–304. <http://dx.doi.org/10.1126/science.227.4684.302>.
- Lemasquerier, D.G., Bierson, C.J., Soderlund, K.M., 2023. Europa's ocean translates interior tidal heating patterns to the ice-ocean boundary. *AGU Adv.* 4, e2023AV000994. <http://dx.doi.org/10.1029/2023AV000994>.

- Levin, S., Zhang, Z., Bolton, S., Brown, S., Ermakov, A., Feng, J., Hartogh, P., Misra, S., Siegler, M., Steffes, P., Stevenson, D., 2024. Results from Juno microwave radiometer observations of Europa's sub-surface ice. DPS2024. URL <https://submissions.miramart.com/DPS56/Itinerary/PresentationDetail.aspx?evdid=63>.
- Manoj, C., Kuvshinov, A., Maus, S., Lühr, H., 2006. Ocean circulation generated magnetic signals. *Earth Planet. Space* 58 (4), 429–437. <http://dx.doi.org/10.1186/BF03351939>.
- Marshall, J., Hill, C., Perelman, L., Adcroft, A., 1997. Hydrostatic, quasi-hydrostatic, and nonhydrostatic ocean modeling. *J. Geophys. Res.: Oceans* 102, 5733–5752. <http://dx.doi.org/10.1029/96JC02776>.
- Marusiak, A.G., Vance, S.D., Panning, M.P., Běhouňková, M., Byrne, P.K., Choblet, G., Daswani, M.M., Hughson, K., Journaux, B., Lobo, A.H., Schmidt, B.E., Sládková, K.P., Soderlund, K.M., Song, W., Souček, O., Steinbrügge, G., Thompson, A.F., Wang, S., 2021. Exploration of icy ocean worlds using geophysical approaches. *Planet. Sci. J.* 2 (4), 150. <http://dx.doi.org/10.3847/PSJ/ac1272>.
- Naseem, M., Neveu, M., Howell, S., Lesage, E., Daswani, M.M., Vance, S.D., 2023. Salt distribution from freezing intrusions in ice shells on ocean worlds: Application to Europa. *Planet. Sci. J.* 4 (9), 181. <http://dx.doi.org/10.3847/PSJ/ace5a2>.
- Petrenko, V.F., Schulson, E.M., 1992. The effect of static electric fields on protonic conductivity of ice single crystals. *Phil. Mag. B* 66 (3), 341–353. <http://dx.doi.org/10.1080/13642819208207654>.
- Petricca, F., Genova, A., Castillo-Rogez, J.C., Styczinski, M.J., Cochrane, C.J., Vance, S.D., 2023. Characterization of icy moon hydrospheres through joint inversion of gravity and magnetic field measurements. *Geophys. Res. Lett.* 50, e2023GL104016. <http://dx.doi.org/10.1029/2023GL104016>.
- Psarakis, C.A., Fidelis, T.T., Chin, K.B., Journaux, B., Kavner, A., Sarker, P., Styczinski, M.J., Vance, S.D., Wei, T., 2024. Electrical conductivity of subsurface ocean analogue solutions from molecular dynamics simulations. *ACS Earth Space Chem.* 8 (6), 1146–1153.
- Roberts, J.H., McKinnon, W.B., Elder, C.M., Tobie, G., Biersteker, J.B., Young, D., Park, R.S., Steinbrügge, G., Nimmo, F., Howell, S.M., Castillo-Rogez, J.C., Cable, M.L., Abrahams, J.N., Bland, M.T., Chivers, C., Cochrane, C.J., Dombard, A.J., Ernst, C., Genova, A., Gerekos, C., Glein, C., Harris, C.D., Hay, H.C.F.C., Hayne, P.O., Hedman, M., Hussmann, H., Jia, X., Khurana, K.K., Kiefer, W.S., Kirk, R., Kivelson, M., Lawrence, J., Leonard, E.J., Lunine, J.I., Mazarico, E., McCord, T.B., McEwen, A., Paty, C., Quick, L.C., Raymond, C.A., Retherford, K.D., Roth, L., Rymer, A., Saur, J., Scanlan, K., Schroeder, D.M., Senske, D.A., Shao, W., Soderlund, K.M., Spiers, E., Styczinski, M.J., Tortora, P., Vance, S.D., Villarreal, M.N., Weiss, B.P., Westlake, J.H., Withers, P., Wolfenbarger, N., Buratti, B., Korth, H., Pappalardo, R.T., Group, The Interior Thematic Working, 2023. Exploring the interior of Europa with the Europa Clipper. *Space Sci. Rev.* 219 (6), 46. <http://dx.doi.org/10.1007/s11214-023-00990-y>.
- Sabaka, T.J., Toffner-Clausen, L., Olsen, N., Finlay, C.C., 2020. CM6: a comprehensive geomagnetic field model derived from both CHAMP and Swarm satellite observations. *Earth Planet. Space* 72, 80. <http://dx.doi.org/10.1186/s40623-020-01210-5>.
- Šachl, L., Knopp, O., Velínský, J., 2024. Electrical conductivity of the suboceanic upper mantle constrained by satellite-derived tidal magnetic fields: 3-D inversion, validation and interpretation. *Geophys. J. Int.* 238, 1254–1268. <http://dx.doi.org/10.1093/gji/ggae209>.
- Šachl, L., Martinec, Z., Velínský, J., Irrgang, C., Petereit, J., Saynisch, J., Einspigel, D., Schnepf, N.R., 2019. Modelling of electromagnetic signatures of global ocean circulation: Physical approximations and numerical issues. *Earth Planet. Space* 71 (1), 58. <http://dx.doi.org/10.1186/s40623-019-1033-7>.
- Šachl, L., Velínský, J., Fullea, J., Martinec, Z., 2022. Inversion of the satellite observations of the tidally induced magnetic field in terms of 3-D upper-mantle electrical conductivity: Method and synthetic tests. *Geophys. J. Int.* 229 (3), 2115–2132. <http://dx.doi.org/10.1093/gji/ggac015>.
- Sanford, T.B., 1971. Motionally induced electric and magnetic fields in the sea. *J. Geophys. Res.* 76 (15), 3476–3492. <http://dx.doi.org/10.1029/JC076i015p03476>.
- Sanford, T.B., 1982. Temperature transport and motional induction in the Florida current. *J. Mar. Res.* 40, 621–639.
- Semtner, A.J., 1974. An Oceanic General Circulation Model with Bottom Topography. Technical Report No. 9, UCLA Department of Meteorology.
- Seufert, M., Saur, J., Neubauer, F.M., 2011. Multi-frequency electromagnetic sounding of the Galilean moons. *Icarus* 214 (2), 477–494.
- Soderlund, K.M., 2019. Ocean dynamics of outer solar system satellites. *Geophys. Res. Lett.* 46, 8700–8710. <http://dx.doi.org/10.1029/2018GL081880>.
- Soderlund, K.M., Schmidt, B., Wicht, J., 2014. Ocean-driven heating of Europa's icy shell at low latitudes. *Nat. Geosci.* 7, 16–19. <http://dx.doi.org/10.1038/ngeo2021>.
- Sohl, F., Spohn, T., Breuer, D., Nagel, K., 2002. Implications from Galileo observations on the interior structure and chemistry of the Galilean satellites. *Icarus* 157 (1), 104–119. <http://dx.doi.org/10.1006/icar.2002.6828>.
- Stephenson, D., Bryan, K., 1992. Large-scale electric and magnetic fields generated by the oceans. *J. Geophys. Res.: Oceans* 97 (C10), 15467–15480. <http://dx.doi.org/10.1029/92JC01400>.
- Styczinski, M.J., Vance, S.D., Daswani, M.M., 2023. PlanetProfile: Self-consistent interior structure modeling for ocean worlds and rocky dwarf planets in Python. *Earth Space Sci.* 10 (8), e2022EA002748. <http://dx.doi.org/10.1029/2022EA002748>.
- Terra-Nova, F., Amit, H., Choblet, G., Tobie, G., Bouffard, M., Čadež, O., 2023. The influence of heterogeneous seafloor heat flux on the cooling patterns of Ganymede's and Titan's subsurface oceans. *Icarus* 389, 115232. <http://dx.doi.org/10.1016/j.icarus.2022.115232>.
- Tyler, R.H., Maus, S., Luhr, H., 2003. Satellite observations of magnetic fields due to ocean tidal flow. *Science* 299 (5604), 239–241. <http://dx.doi.org/10.1126/science.1078074>.
- Van Hoolst, T., Tobie, G., Vallat, C., Altobelli, N., Bruzzone, L., Cao, H., Dirx, D., Genova, A., Hussmann, H., Iess, L., Kimura, J., Khurana, K.K., Lucchetti, A., Mitri, G., Moore, W., Saur, J., Stark, A., Vorbürger, A., Wieczorek, M., Aboudan, A., Bergman, J., Bovolenta, F., Breuer, D., Cappuccio, P., Carrer, L., Ceconci, B., Choblet, G., De Marchi, F., Fayolle, M., Fienga, A., Futaana, Y., Hauber, E., Kofman, W., Kumamoto, A., Lainey, V., Molyneux, P., Mousis, O., Plaut, J., Puccio, W., Retherford, K., Roth, L., Seignovart, B., Steinbrügge, G., Thakur, S., Tortora, P., Tosi, F., Zannoni, M., Barabash, S., Dougherty, M., Gladstone, R., Gurvits, L.I., Hartogh, P., Palumbo, P., Poulet, F., Wahlund, J.-E., Grasset, O., Witasse, O., 2024. Geophysical characterization of the interiors of Ganymede, Callisto and Europa by ESA's Jupiter ICy moons explorer. *Space Sci. Rev.* 220 (5), 54. <http://dx.doi.org/10.1007/s11214-024-01085-y>.
- Vance, S.D., Craft, K.L., Shock, E., Schmidt, B.E., Lunine, J., Hand, K.P., McKinnon, W.B., Spiers, E.M., Chivers, C., Lawrence, J.D., Wolfenbarger, N., Leonard, E.J., Robinson, K.J., Styczinski, M.J., Persaud, D.M., Steinbrügge, G., Zolotov, M.Y., Quick, L.C., Scully, J.E. C., Becker, T.M., Howell, S.M., Clark, R.N., Dombard, A.J., Glein, C.R., Mousis, O., Sephton, M.A., Castillo-Rogez, J., Nimmo, F., McEwen, A.S., Gudipati, M.S., Jun, I., Jia, X., Postberg, F., Soderlund, K.M., Elder, C.M., 2023. Investigating Europa's habitability with the Europa Clipper. *Space Sci. Rev.* 219 (8), 81. <http://dx.doi.org/10.1007/s11214-023-01025-2>.
- Vance, S.D., Panning, M.P., Stähler, S., Cammarano, F., Bills, B.G., Tobie, G., Kamata, S., Kedar, S., Sotin, C., Pike, W.T., Lorenz, R., Huang, H.-H., Jackson, J.M., Banerdt, B., 2018. Geophysical investigations of habitability in ice-covered ocean worlds. *J. Geophys. Res.: Planets* 123, 180–205. <http://dx.doi.org/10.1002/2017JE005341>.
- Vance, S.D., Styczinski, M.J., Bills, B.G., Cochrane, C.J., Soderlund, K.M., Gómez-Pérez, N., Paty, C., 2021. Magnetic induction responses of Jupiter's ocean moons including effects from adiabatic convection. *J. Geophys. Res.: Planets* 126, e2020JE006418. <http://dx.doi.org/10.1029/2020JE006418>.
- Varshalovich, D.A., Moskalev, A.N., Khersonskii, V.K., 1988. Quantum Theory of Angular Momentum. World Scientific, Singapore. <http://dx.doi.org/10.1142/0270>.
- Velínský, J., Grayver, A., Kuvshinov, A., Šachl, L., 2018. On the modelling of M_2 tidal magnetic signatures: Effects of physical approximation and numerical resolution. *Earth Planet. Space* 70 (1), 70–192. <http://dx.doi.org/10.1186/s40623-018-0967-5>.
- Velínský, J., Martinec, Z., 2005. Time-domain, spherical harmonic-finite element approach to transient three-dimensional geomagnetic induction in a spherical heterogeneous Earth. *Geophys. J. Int.* 160, 81–101. <http://dx.doi.org/10.1111/j.1365-246X.2005.02546.x>.
- Velínský, J., Šachl, L., Martinec, Z., 2019. The global toroidal magnetic field generated in the Earth's oceans. *Earth Planet. Sci. Lett.* 509, 47–54. <http://dx.doi.org/10.1016/j.epsl.2018.12.026>.
- Velínský, J., Schnepf, N.R., Nair, M.C., Thomas, N.P., 2021. Can seafloor voltage cables be used to study large-scale circulation? An investigation in the Pacific Ocean. *Ocean Sci.* 17 (1), 383–392. <http://dx.doi.org/10.5194/os-17-383-2021>.
- Visbeck, M., Marshall, J., Haine, T., Spall, M., 1997. Specification of eddy transfer coefficients in coarse-resolution ocean circulation models. *J. Phys. Oceanogr.* 27, 381–402. [http://dx.doi.org/10.1175/1520-0485\(1997\)027<0381:SOETCI>2.0.CO;2](http://dx.doi.org/10.1175/1520-0485(1997)027<0381:SOETCI>2.0.CO;2).
- Vivier, F., Maier-Reimer, E., Tyler, R.H., 2004. Simulations of magnetic fields generated by the Antarctic Circumpolar Current at satellite altitude: Can geomagnetic measurements be used to monitor the flow? *Geophys. Res. Lett.* 31 (10), <http://dx.doi.org/10.1029/2004GL019804>.
- Zeng, Y., Jansen, M.F., 2021. Ocean circulation on Enceladus with a high- versus low-salinity ocean. *Planet. Sci. J.* 2, 151. <http://dx.doi.org/10.3847/PSJ/ac1114>.
- Zimmer, C., Khurana, K.K., Kivelson, M.G., 2000. Subsurface oceans on Europa and Callisto: Constraints from Galileo magnetometer observations. *Icarus* 147 (2), 329–347. <http://dx.doi.org/10.1006/icar.2000.6456>.